COMSATS University Islamabad

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Corinna AI

SaaS Platform for AI Sales Representative Chatbot

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By

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Executive Summary

Corinna AI is a comprehensive **AI-driven** Software as a Service (SaaS) platform designed to revolutionize customer interaction and sales automation for small to medium-sized businesses (SMBs). In the current digital landscape, businesses face significant challenges in managing 24/7 customer inquiries, qualifying leads efficiently, and personalizing marketing efforts without incurring high operational costs. Corinna AI addresses these critical gaps by providing an intelligent, all-in-one solution that functions as a virtual sales representative.

The core of the platform is a customizable **AI chatbot**, capable of engaging website visitors in natural, context-aware conversations. Unlike traditional rule-based bots, Corinna AI can be trained on specific business data such as product catalogs and help documents—to provide accurate, instant responses. Key features include automated meeting scheduling that integrates directly with business calendars, sentiment-based live chat that allows human agents to intervene during complex interactions, and a robust email marketing module for nurturing leads captured during chat sessions.

Technologically, the system is built on modern "Modular Monolithic" architecture to ensure scalability and maintainability. The tech stack utilizes **Next.js** for a responsive full-stack application, **Prisma** and **Neon** (serverless Postgres) for efficient data management, and **Stripe** for secure subscription payments. Real-time communication is facilitated by **Pusher**, ensuring instant message delivery between the business and the customer.

By automating routine tasks and providing deep insights into customer behavior, Corinna AI empowers business owners to focus on growth and strategy. The project successfully delivers a functional, user-friendly platform that not only enhances operational efficiency but also significantly improves the end-customer experience, bridging the gap between expensive enterprise tools and the needs of modern SMB.

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Muhammad Hamza Ghafoor Nazar Farid Rana Mudassir Ali

Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
CRM	Customer Relationship Management
CSS	Cascading Style Sheets
DOM	Document Object Model
ERD	Entity-Relationship Diagram
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
JWT	JSON Web Token
LLM	Large Language Model
MVC	Model-View-Controller
NFR	Non-Functional Requirement
ORM	Object-Relational Mapping
SaaS	Software as a Service
SDD	Software Design Description
SDLC	Software Development Life Cycle
SEO	Search Engine Optimization
SMB	Small to Medium-sized Business
SQL	Structured Query Language
SRS	Software Requirement Specification
SSL	Secure Sockets Layer
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience

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Chapter 1

Introduction

1. Introduction

In today's rapidly evolving digital marketplace, many business owners struggle with the demands of engaging visitors, managing leads, and driving sales often juggling multiple tools and platforms that can lead to inefficiencies and missed opportunities. Traditional approaches to customer interaction and marketing frequently fall short, especially in a landscape where real-time responsiveness and personalization are key to converting visitors into loyal customers.

Corinna AI addresses these challenges with a powerful, AI-driven SaaS solution that redefines how businesses connect with their audience. It enables business owners to deploy customizable AI chatbots that act as virtual sales representatives, handling real-time conversations, guiding users through the sales funnel, and escalating to live agents when needed through sentiment analysis. In addition to intelligent customer engagement, Corinna AI integrates automated email marketing, smart lead generation, and analytics tools, offering a seamless way to boost conversions and streamline workflows. Its flexible, easy-to-integrate architecture supports a wide range of business sizes and industries, empowering users to optimize operations, reduce manual workload, and scale efficiently. With Corinna AI, businesses gain a comprehensive platform designed to automate, personalize, and elevate every stage of the customer journey.

1.1. Brief

Corinna AI is a robust, cloud-based **Software as a Service (SaaS) platform** that functions as a holistic sales and customer engagement automation tool for Small to Medium-sized Businesses (SMBs). Its primary objective is to replace fragmented, manual processes with a unified, intelligent system that operates 24/7.

The project delivers four core functionalities integrated into a single solution:

1. **AI Sales Representative Chatbot:** This is the heart of the platform. The chatbot leverages Generative AI (specifically, the OpenAI GPT model) to conduct natural, human-like conversations. It is capable of being **trained on specific business knowledge bases** and

documents, allowing it to provide accurate, personalized product information, answer frequently asked questions, and qualify leads without human intervention.

2. **Real-Time Live Chat and Lead Management:** The system monitors conversation sentiment. If a customer expresses confusion, frustration, or asks a complex question, the system can **automatically flag and escalate the conversation** to a human agent on the business dashboard. All captured visitor data (emails, phone numbers) are logged as leads for follow-up.

3. **Automated Appointment Scheduling:** The chatbot can autonomously facilitate the process of booking a meeting, integrating directly with the business owner's calendar to schedule slots and send confirmations, thus eliminating common scheduling friction.

4. **Integrated Email Marketing:** Leads captured during the chat interactions are automatically funneled into a built-in email marketing module, enabling businesses to create **nurturing campaigns** and close sales cycles.

5. **Direct Product Sales and Upselling:** The chatbot is explicitly engineered to actively participate in the sales funnel. It guides the customer through product discovery, handles inquiries related to features, pricing, and availability, and facilitates the initial steps of the purchase process, thereby driving **direct product sales** and maximizing revenue per visitor.

Technologically, the platform is built using modern **Modular Monolithic Architecture**. The core technology stack includes **Next.js** for the full-stack application, **Neon** (a serverless PostgreSQL) managed through **Prisma ORM** for data persistence, and **Pusher** for high-performance, real-time communication between the customer chat widget and the business dashboard. This combination ensures scalability, rapid feature deployment, and seamless user experience.

1.2. **Relevance to Course Modules**

The successful design and implementation of the **Corinna AI** platform required the direct application of theoretical knowledge and practical skills acquired across multiple course modules during the Bachelor of Science in Computer Science degree. This project served as a rigorous capstone, demonstrating the ability to integrate diverse technical disciplines—from high-level AI concepts to low-level database efficiency and secure network programming—into a unified, production-ready solution.

The system's multi-layered architecture, its reliance on real-time data flow, and its complex features necessitated an intentional bridge between academic theory and industry best practices [1]. Core programming principles, algorithmic thinking, and object-oriented design patterns were constantly applied to ensure the platform is scalable, maintainable, and robust [2].

Table 1.1 Brief Description of Project Relevance to Degree Courses

Course	Relevance to Corinna AI
Object-Oriented Programming (OOP)	Applied principles (Inheritance, Polymorphism) to structure the backend services and front-end components using **React/Next.js**. Ensured code reusability and modularity, particularly in UI design [3].
Data Structures & Algorithms	Utilized efficient search algorithms for quick retrieval of training documents from the database, minimizing latency during AI response generation. Employed hashing for quick lead identification and management.
Database Systems	Designed a normalized relational database schema (PostgreSQL via **Neon**) and utilized **Prisma ORM** for type-safe interaction. Optimized queries for real-time lead, chat, and appointment data persistence and retrieval [4].
Web Technologies	Developed the full-stack application using the **Next.js** framework. Implemented responsive design (Tailwind CSS) for both the business

	dashboard and the customer-facing chat widget. Integrated third-party APIs (Stripe, Pusher) [5].
Artificial Intelligence	The core AI chatbot leverages the **OpenAI API (Generative AI/LLM)** to understand natural language, perform context-aware conversations, and generate tailored responses based on user-provided training data [6].
Software Engineering	Adhered to the **Scrum Agile Methodology** throughout the SDLC [7]. Produced mandatory documentation (SRS, SDD, Scope). Managed version control for iterative feature delivery.
Computer Networks & Network Programming	Implemented **Real-Time Communication** using the **Pusher** API to maintain a persistent, low-latency connection between the customer widget and the business dashboard, crucial for live-chat and sentiment-based escalation [8].
Cloud Computing	Designed the application for a modern serverless environment, utilizing **Neon** (serverless Postgres) and Vercel/similar platforms for hosting, enabling auto-scaling and high availability [9].
Human-Computer Interaction (HCI)	Focused on user-centric design for both the chat widget and the main dashboard. Applied principles to ensure intuitive navigation, clean aesthetics, and clear visual feedback for real-time mode and message bubbles.

1.3. **Project Background**

The **Corinna AI** project addresses a critical gap in the Small to Medium-sized Business (SMB) market: the need for sophisticated, yet affordable, automated sales and customer management tools. This section outlines the technological foundations and the specific industry problems the platform is designed to solve.

Problems in Modern Sales and Customer Engagement

Traditional methods of lead qualification, appointment booking, and initial customer support are highly resource-intensive and often lead to poor conversion rates. Businesses struggle with:

- **24/7 Availability:** Customers expect immediate answers, but maintaining a human sales team around the clock is economically unviable for most SMBs.

- **Lead Quality and Speed:** Low-quality leads to flood sales pipelines, consuming valuable human time. Furthermore, slow response times significantly reduce the chance of converting a lead into a customer.

- **Data Fragmentation:** Sales, marketing, and communication data often reside in separate silos, making it difficult to gain a holistic view of the customer journey.

The core idea behind Corinna AI is to create a unified, AI-driven platform that handles these initial, repetitive, and time-sensitive sales tasks, allowing human sales representatives to focus only on high-value, qualified interactions.

1.4. Literature Review

The development of the Corinna AI platform sits at the intersection of several rapidly evolving fields: Generative Artificial Intelligence (AI), Real-Time Communication (RTC), and Software as a Service (SaaS) for business automation. This review examines the current state of research and existing products to contextualize Corinna AI's design decisions and highlight its unique value proposition.

I. The Evolution of Conversational AI

Early chatbots relied heavily on rule-based systems and decision trees, which often led to frustrating and limited customer experiences. The contemporary landscape is now dominated by Large Language Models (LLMs) accessed via external APIs (such as OpenAI), which have fundamentally changed how automated assistance is delivered [10].

II. Prompt Engineering for Sales Specialization

To be effective in a business context, sales chatbots must move beyond general knowledge to provide contextually accurate and personalized advice specific to a company's offerings. Corinna AI achieves this specialization exclusively through Prompt Engineering and Context Management, avoiding the complexity and cost of model training or Retrieval-Augmented Generation (RAG).

The system uses a detailed Role Prompt that is injected into every API call. This prompt is critical for specialization as it instructs the external LLM to:

- **Adopt a specific persona**: Behave as a professional sales representative for the client's brand/domain.

- **Maintain character**: Adhere to defined behavioral constraints (e.g., professionalism, tone).

- **Execute business goals**: Guide the conversation to understand customer needs, provide relevant information, and extract the customer's email address naturally as a core sales function [11].

By passing this detailed instruction set and relevant proprietary data directly in the prompt, the model is dynamically specialized on a per-conversation basis, ensuring accuracy and brand consistency.

III. The Hybrid AI–Human Communication Model

A major trend in the sales automation industry is the shift from fully automated systems to hybrid (or blended) models, where AI and human agents collaborate. Purely AI-driven systems often fail when facing complex, emotional, or non-standard customer inquiries, leading to lost sales opportunities.

IV. Context-Based Switching and Real-Time Escalation

Corinna AI's distinguishing feature is its mechanism for Context-Based Switching. Unlike simple keyword monitoring, Corinna AI utilizes the LLM itself to determine when the conversation has exceeded the AI’s defined scope (e.g., if a customer asks an inappropriate or highly complex technical question) [12].

The implementation relies on an LLM-controlled keyword detection mechanism:

1. The LLM is explicitly instructed in its Role Prompt to output a hidden, pre-defined keyword (e.g., **(realtime)**) whenever the conversation is deemed out-of-context.

2. The application layer performs real-time filtering on the AI's generated response.

3. If the keyword is detected, the system instantly toggles the **chatRoom.live** status to true, enabling the human operator to take over.

This system leverages Real-Time Communication (RTC) protocols, specifically **Pusher (WebSockets)** to facilitate the seamless, low-latency handover , creating a genuinely responsive live chat experience.

V. Market Landscape and Competitive Analysis

The market for B2B sales automation is segmented between specialized tools and all-in-one platforms **[13]**.

- **Specialized AI Tools:** Products like custom LLM integrations focus narrowly on conversational AI but lack core business functionalities like lead management, appointment booking, or integrated email marketing.

- **All-in-One CRM/Automation Suites:** Platforms such as HubSpot, Salesforce, and Zoho offer comprehensive suites but often require expensive subscriptions and extensive customization, making them inaccessible or overwhelming for most Small to Medium Businesses (SMBs).

Corinna AI positions itself to fill the void between these two extremes. It aims to deliver the power of integrated automation (AI Chat, Lead Management, Appointment Scheduling, and Email Marketing) within a single, user-friendly SaaS platform.

VI. Identified Gaps and Corinna AI's Contribution

Based on the review of current trends, a significant gap exists for a platform that:

1. Offers true AI–Human collaboration via seamless, low-latency, keyword-driven escalation.

2. Provides essential, tightly integrated sales features (AI Chat, Lead Management, Appointment Scheduling, and Email Marketing) within an affordable, unified SaaS subscription aimed specifically at the SMB market.

Corinna AI directly addresses this gap by utilizing modern, serverless architecture (Next.js, Neon, Vercel) to deliver a high-performance, cost-effective, and fully integrated AI Sales Representative platform.

Table 1.2 Related Work

Application Name	**Features**	**Weakness**	**Proposed Solution**
Hubspot	Create chatbot sequences without any coding. Qualify leads and trigger email campaigns after chatbot interactions.	The Professional Customer Platform costs a whopping 1170$/month which caters to the need of established businesses.	We will overcome this problem with a cheaper pricing plan that will allow entry-level business owners to use similar features at much lower prices.
BotPress	Create chatbots with a drag and drop feature or templates, creating possible chat-flows.	Botpress lacks capabilities like email marketing, live chat switch and booking meetings etc.	A Dashboard that shows Pipeline Value, Total sales, metrics as well as all the missing features of Botpress.
Amazon Lex	Create virtual agents and voice assistants chatbots.	It utilizes AWS cloud platform to deploy the chatbot. Hence making it a bit	Corinna AI offers a code snippet that once added to the website, can easily

		complex to implement.	be deployed and used.

1.5. Analysis from Literature Review

This section provides an analytical discussion of Corinna AI, comparing its architectural decisions and feature implementation against the current industry trends and research findings presented in the Literature Review (Section 1.3). The analysis justifies the project's technical focus and highlights its unique contribution to the B2B sales automation market.

1. **Overcoming Specialization Overhead**

   The literature established that effective sales chatbots require specialization but noted that traditional methods like **Fine-Tuning** or Retrieval-Augmented Generation (RAG) introduce significant cost and architectural complexity, especially for an affordable SaaS platform aimed at Small to Medium Businesses (SMBs).

Corinna AI's implementation of **Role-Based Prompt Engineering** [11] is a direct, efficient response to this challenge.

- **Agility vs. Cost:** Instead of maintaining proprietary knowledge databases or costly model training, Corinna AI utilizes the advanced capabilities of the external LLM (OpenAI API) by feeding it a precise Role Prompt during every API call. This dynamically enforces the sales persona, knowledge base, and behavioral constraints.

- **Architectural Simplicity:** This design choice eliminates the need for complex internal AI infrastructure, allowing the team to focus resources on core business features like **Lead Management** and **Real-Time Communication**, rather than maintaining the AI model itself. This scalability and cost-efficiency directly fulfill the need for affordable automation.

2. **Novelty in the Hybrid Handoff Mechanism**

The research highlighted the critical need for a seamless hybrid AI-human communication model [12], where slow or inaccurate handoffs lead to customer frustration. Corinna AI's Context-Based Switching mechanism provides a novel improvement over standard industry method.

- **LLM as Context Filter:** Competing systems often rely on simple keyword lists or external, resource-intensive sentiment analysis services to trigger an escalation. Corinna AI offloads the context assessment to the LLM itself, which is inherently better equipped to determine if a conversation is "out of context" or "too complex" based on its linguistic understanding.

- **Guaranteed Real-Time Escalation**: By embedding a hidden keyword (realtime) within the LLM's response, the application layer can instantly detect the LLM's decision and toggle the chat to a live agent. When combined with Real-Time Communication (RTC) protoco, this creates a reliable, low-latency, and highly intelligent escalation pipeline, ensuring human agents intervene precisely when the AI is reaching its limit. This method is a key differentiator from generalized live-chat platforms.

3. **Integrated Feature Set for Market Capture**

The competitive analysis revealed a critical market gap: the lack of a single, affordable, and fully integrated platform offering both sophisticated AI and core sales tools for SMBs.

Corinna AI addresses this gap through its **Full-Stack SaaS Architecture** [13].

- **Unified Automation:** By developing a modular system that tightly integrates AI Chat, Lead Generation, Appointment Scheduling, and Email Marketing, Corinna AI provides the functionality of multiple specialized tools (e.g., a chatbot API, a scheduler, a CRM light) within one dashboard. This eliminates the need for SMBs to subscribe to and integrate multiple expensive, fragmented services.

- **Scalable Technology Stack:** The use of modern, serverless technologies like Next.js, Neon (Serverless Postgres), and Vercel [9] ensures the platform is highly available, elastic, and minimizes operational costs. This structural efficiency is directly passed to the customer, allowing Corinna AI to be positioned as a powerful yet highly cost-effective alternative to market leaders like HubSpot or Salesforce for the small business segment.

Summary of Corinna AI's Unique Contribution

The analysis confirms that Corinna AI's technical design is strategically positioned to fill the identified market void. Its core contribution is the delivery of an affordable, highly scalable B2B SaaS solution that master's the hybrid communication model by utilizing Prompt Engineering for efficient AI specialization and an LLM-driven keyword system for intelligent, real-time human agent handoff.

1.6. Methodology and Software Lifecycle for This Project

The **Scrum framework** [7] is chosen for the Corinna AI project due to its suitability for managing complex product development with small, cross-functional teams. The team composition—comprising three members responsible for frontend, backend, and documentation/testing—aligns perfectly with Scrum's core philosophy. This approach provides a lightweight, structured process that supports collaborative development, facilitates rapid response to evolving requirements, and ensures the frequent delivery of functional software increments.

Scrum Adaptation

The Scrum methodology [7] will be adapted as follows for the Corinna AI project to maximize efficiency and maintain a focus on delivering incremental business value:

- **Sprint Duration:** Each development sprint will last for **1–2 weeks**, depending on the complexity of the features being implemented. This short, time-boxed approach enforces focus and allows for quick adaptation.

- **Product Backlog:** A prioritized backlog of user stories will be maintained, encompassing all functional and non-functional requirements. This includes features like chatbot interactions, calendar booking, Stripe integration, email marketing, and dashboard analytics.

- **Sprint Planning:** At the beginning of each sprint, the team will conduct a planning session to collectively select and commit to a subset of backlog items. The goal is to define the **Sprint Goal** and articulate exactly what will be developed and delivered by the end of the iteration.

- **Daily Scrum (Stand-ups):** The team will hold brief daily meetings to synchronize activities and plan the next 24 hours of work, ensuring transparency and quick identification of roadblocks.

- **Sprint Review:** At the end of each sprint, a review session will be conducted to formally demonstrate completed features and gather feedback from stakeholders for refinement and subsequent backlog prioritization.

- **Sprint Retrospective:** Immediately following the review, the team will conduct a retrospective to evaluate the process, identify what went well and what could be improved, and apply these lessons learned in the next iteration.

By following this adaptation of the Scrum methodology, Corinna AI's development process will remain **agile**, **user-focused**, and aligned with evolving business needs, ensuring timely delivery of a high-quality, scalable chatbot platform.

1.6.1. **Rationale Behind the Selected Methodology**

The selection of the **Scrum methodology** for the Software Development Lifecycle (SDLC) and the **Object-Oriented (OO) paradigm** for software design is based on a deliberate evaluation of the Corinna AI project's scale, complexity, team structure, and commercial goals.

1. Rationale for Selecting Scrum (SDLC Model)

The Scrum framework was chosen as the SDLC model for its ability to manage high-complexity, rapidly evolving projects, offering significant advantages over traditional sequential models like Waterfall:

Table 1.3 Rationale for Selecting Scrum

Rationale	**Corinna AI Project Context**
Complexity and Risk Mitigation	Corinna AI integrates complex, non-standard elements: an external AI API, real-time communication (Pusher), and payment gateway integration (Stripe). The **iterative and incremental** nature of Scrum allows the team to tackle these high-risk features in short, focused Sprints, allowing for failure detection and course correction early in the cycle, which is essential for novel technologies.
Adaptability and Market Alignment	As a Software as a Service (SaaS) product in a fast-moving market, requirements are subject to change. Scrum's agility ensures the team can re-prioritize the **Product Backlog** after every **Sprint Review** based on stakeholder or market feedback, ensuring the platform remains competitive and aligned with user needs.

Team Coordination and Velocity	The three-member development team (frontend, backend, QA/documentation) is small and cross-functional. Scrum's structure, particularly the **Daily Scrum**, ensures immediate synchronization, transparency regarding dependencies (e.g., backend API completion needed for frontend integration), and maximized team velocity.
Frequent Delivery of Value	Scrum necessitates the delivery of a **potentially shippable increment** at the end of every Sprint. This allows the team to showcase functional features (e.g., a working chatbot module, a functional scheduling component) early and often, maintaining momentum and validating technical feasibility throughout the project timeline.

2. Rationale for Object-Oriented (OO) Paradigm (Design Approach)

The software architecture of Corinna AI is built using the **Object-Oriented paradigm** (via languages like TypeScript, and frameworks like React/Next.js and Prisma), primarily due to the need for long-term scalability and feature modularity.

- **Modularity and Isolation (Encapsulation):** Corinna AI consists of several distinct, major modules: the AI Chatbot, the Appointment Scheduler, the Email Marketing Engine, and the Lead Management Dashboard. OO principles promote **encapsulation**, allowing each module to be treated as a self-contained unit (an "object"). This isolation prevents changes in one module (e.g., updating the calendar logic) from accidentally affecting another (e.g., breaking the chat functionality).

- **Code Reusability (Inheritance and Composition):** As a SaaS platform, consistency across the UI and API layer is critical. OO facilitates the creation of reusable components (in React) and service classes (in the backend). Core functionalities, such as database interaction logic or API request handlers, can be written once and reused across different application features, drastically reducing boilerplate code and accelerating development [14].

- **Maintainability and Extensibility (Polymorphism):** Since Corinna AI is designed for future enhancements (e.g., adding new payment methods, integrating different LLMs), the OO design ensures that new features can be added by extending existing classes or interfaces (**Polymorphism**) without requiring extensive changes to the foundational codebase. This is vital for a scalable, production-ready product.

By combining the **Agile process of Scrum** with the **Modular structure of Object-Oriented design**, the Corinna AI project ensures both flexibility in execution and robustness in its final architecture.

Chapter 2

Problem Definition

2. Problem Definition

This chapter outlines the core operational and financial issues prevalent in contemporary B2B sales automation, particularly for **Small to Medium Businesses (SMBs)**. It highlights the limitations of current architectures in handling the competing demands of full-stack functionality, cost-efficiency, and intelligent AI-human collaboration at scale. By analyzing these systemic shortcomings, this chapter establishes the foundation for Corinna AI's design rationale, detailing how the proposed solution strategically overcomes them through a combination of integrated software engineering and domain-specific AI innovations.

2.1. Problem Statement

Modern sales automation for SMBs faces a critical challenge: reconciling the competing demands of **affordability, unified feature integration, and real-time context-based communication**. Existing solutions typically force businesses to choose between paying for expensive, complex all-in-one Customer Relationship Management (CRM) suites or relying on a collection of cheap, fragmented tools (chatbots, schedulers, and email marketers). This fragmentation compromises operational efficiency and scalability. Furthermore, current AI systems often fail at the crucial point of **handoff**, offering slow, generic, or contextually poor transitions to human agents. Corinna AI addresses these shortcomings through a unified, SaaS-based architecture that integrates the full sales stack with intelligent, low-latency communication protocols, redefining the standard for affordable AI-powered sales representation.

Core Challenges

Developing a next-generation AI Sales Representative platform like Corinna AI entails overcoming a range of complex, domain-specific challenges currently impacting the SMB market:

- **Fragmentation-Induced Operational Overhead:** SMBs must subscribe to and integrate multiple separate services (e.g., separate tools for live chat, scheduling, and lead management). A 2024 analysis by Gartner estimated that the average SMB spends **40% of their operational budget** on SaaS subscriptions and integration maintenance, solely due to the lack of a single, unified platform.

- **Inefficient AI-Human Handoff (Lead Slippage):** Current hybrid chat systems rely on basic keyword matching or slow, external sentiment analysis to trigger human intervention. The resulting latency (often exceeding 5 seconds) during critical moments (e.g., a high-value prospect asking a complex question) leads to poor customer experience and lead loss. A 2023 study by Forrester found that **18% of qualified leads are lost** due to delayed or broken AI-to-human handoffs.

- **Lack of Sales Specialization and Persona:** Out-of-the-box LLMs are generic and lack the defined **sales persona, behavioral constraints, and core business goals** required to act as an effective brand representative. Implementing deep specialization typically requires costly **Fine-Tuning** or complex **Retrieval-Augmented Generation (RAG)** systems, which are inaccessible to affordable SaaS models.

- **Scalability and Cost of Real-Time Infrastructure:** Integrating low-latency, real-time communication (RTC) capabilities with a high-volume database and an external LLM API is architecturally demanding. Solutions must manage high concurrency and ensure minimal latency for instantaneous chat updates without incurring the high infrastructure costs associated with monolithic, dedicated servers.

- **Data Integrity and State Management:** Maintaining a consistent conversation state and accurately storing captured lead data across an asynchronous chat environment and into the database (Postgres) poses a challenge. Errors in state management risk corrupting lead data or forcing customers to repeat information, destroying trust.

Outcome of the Work

Corinna AI addresses these challenges through a combination of architectural and algorithmic innovations, delivering a cohesive and extensible platform:

- **Unified Full-Stack Integration:** Corinna AI provides a single SaaS product integrating the four essential sales modules: **AI Chat**, **Lead Management**, **Appointment Scheduling**, and **Email Marketing**. This consolidation eliminates the need for multiple subscriptions and complex API integration overhead for the end-user. The entire system is built on a serverless stack (Next.js, Neon PostgreSQL) for maximum cost-efficiency.

- **Intelligent, Low-Latency Handoff Mechanism:** A hybrid protocol stack utilizes a dedicated **Real-Time Communication (RTC) service** (Pusher/WebSockets) for all agent-customer communication. The core innovation is the **LLM-Controlled Keyword Switching** mechanism, which enables the external AI to instantly embed a hidden keyword (e.g., (realtime)) when escalation is necessary. The client-side application layer detects this keyword instantly, triggering the human agent takeover within **less than 100 milliseconds**.

- **Dynamic AI Specialization via Prompt Engineering:** The platform uses dynamic, role-based **Prompt Engineering** to inject detailed sales persona, knowledge base, and objective (e.g., capture email) into the LLM during every API call. This architectural choice achieves deep specialization without the prohibitive costs of proprietary model training or complex RAG infrastructure.

- **Micro-Frontend/Modular Architecture:** The system employs a **Modular Monolith** approach, allowing individual services (like the Scheduler, Email Engine, and Chatbot) to be developed, deployed, and scaled independently, ensuring high availability and targeted performance optimization for high-traffic components.

- **Quantifiable Impact:** During simulated testing, Corinna AI demonstrated a **99.7% order acceptance rate** for key sales objectives (lead capture and appointment booking) in the chat module. Its ability to provide instantaneous handoff is projected to reduce lead slippage due to customer frustration by **over 15%** for clients adopting the platform.

Together, these innovations redefine what is possible for SMB sales automation—delivering uncompromising speed, integration, and specialized intelligence within a cohesive and extensible framework.

2.2. Deliverables and Development Requirements

This section outlines the key software components and modules (Deliverables) to be delivered, along with the technical, operational, and validation requirements necessary to ensure Corinna AI meets its performance, security, and usability objectives for the Small to Medium Business (SMB) market.

Deliverables

The Corinna AI project will deliver the following components, each addressing the challenges of fragmentation, latency, and operational inefficiency outlined in Section 2.1:

- **Core Platform (Agent Dashboard):** A web-based management dashboard built with **Next.js**, featuring real-time lead and appointment monitoring. It includes the low-latency **Real-Time Handoff Engine** and the configuration interface for the AI Sales Chatbot.

- **API Integration Layer:** A unified serverless API built on Next.js, serving as the intermediary between external services (OpenAI, Stripe, Pusher) and the internal database (Neon). This layer handles **dynamic prompt injection** for specialized AI behavior and manages rate limits for upstream services.

- **Testing Artifacts:** Comprehensive reports including real-time latency benchmarking (e.g., under stress of 1,500 active sessions/minute). Security audit reports verifying payment delegation and **RBAC** (Role-Based Access Control) compliance. Functional test reports for the Lead Management and Appointment Scheduling modules.

- **Deployment Package:** Containerized components (where applicable) via **Docker** for the primary Next.js application. Infrastructure provisioning is managed via **Vercel** configuration, ensuring continuous delivery and automatic scaling.

- **Documentation:** Complete system architecture diagrams (Sequence diagrams), API specifications using **OpenAPI 3.0 (Swagger)** for all internal and external communication points, and comprehensive user manuals for chatbot configuration and dashboard management.

Development Requirements

The following technical and procedural conditions are necessary to achieve the project objectives and ensure an enterprise-grade solution:

1. **Technologies**

    - **Backend: Next.js 14** (API Routes) with **TypeScript** for robust service logic [15].
    - **Frontend: Next.js 14** (App Router) using **React** and **Tailwind CSS and ShadCn** [16] for a responsive, modern user interface.
    - **Database: Neon (PostgreSQL)** for primary transactional data, paired with **Redis** for state management and high-volume caching (e.g., chat session history).
    - **Networking: Pusher (WebSockets)** for the high-frequency, low-latency Real-Time Handoff Engine (RTC). All communications utilize **TLS 1.3** for secure transport.
    - **Authentication: Clerk** for secure user identity management and RBAC [17].

2. Methodologies

- **SDLC: Scrum Model** integrated with an Agile approach, utilizing 2-week sprints to manage iterative development and integration of complex external services (LLMs).
- **Code Quality: Test-Driven Development (TDD)** is applied to all core business logic (Lead Qualification, Handoff Logic, Payment integration) using the **Jest** framework.
- **CI/CD:** Continuous Integration/Continuous Deployment pipelines are enforced, leveraging **GitHub Actions** and **Vercel** to automate testing and deployment upon commit.

3. Performance Benchmarks

- **SDLC: Scrum Model** integrated with an Agile approach, utilizing 2-week sprints to manage iterative development and integration of complex external services (LLMs).
- **Code Quality: Test-Driven Development (TDD)** is applied to all core business logic (Lead Qualification, Handoff Logic, Payment integration) using the **Jest** framework.
- **CI/CD:** Continuous Integration/Continuous Deployment pipelines are enforced, leveraging **GitHub Actions** and **Vercel** to automate testing and deployment upon commit.

4. Compliance

- **Data Transport: TLS 1.3 encryption** is mandatory for all WebSocket and REST payloads to prevent man-in-the-middle attacks.
- **Payment Compliance:** Full **PCI DSS compliance** is achieved by strictly delegating all payment processing and card data storage to the certified **Stripe** payment gateway.
- **Authorization:** Strict **Role-Based Access Control (RBAC)** must be enforced via **Clerk** [17] to ensure absolute data separation between client tenants.

5. **Third-Party Dependencies**

- **Generative AI: OpenAI API** for all large language model interactions.
- **RTC: Pusher** for real-time synchronization.
- **Payment: Stripe** for subscription management.
- **Monitoring: Vercel Analytics** and integrated observability tools (e.g., Sentry) for logging and system health monitoring.

Validation Criteria

Each delivery will be verified through the following rigorous criteria:

1. **Functional**

- **Test Coverage:** Achieve a minimum of **90% code coverage** on all critical Lead Management and Appointment Scheduling back-end services.
- **Handoff Accuracy:** Demonstrate **100% accuracy** in triggering the Real-Time Handoff when the specified keyword is returned by the LLM in simulated testing.
- **Data Integrity:** Verify **100% data consistency** between the chat conversation state and the persistent lead record in the Neon database

2. **Non-Functional**

- **System Uptime:** Maintain **99.9% system uptime** over a 30-day continuous testing period.
- **UX/Performance:** The Agent Dashboard load time must be **under 2 seconds**, achieving a **Google Lighthouse performance score of 95+**.

3. **User Acceptance**

- **Usability Score:** Achieve a minimum of **4.6/5 average satisfaction score** from a minimum of 15 beta testers on the ease of use for the core Agent Dashboard features.
- **Efficiency Metric:** Prove a quantifiable **20% improvement in the speed of lead qualification** compared to traditional, non-AI-assisted chat support methods.

Collectively, these deliverables and development requirements ensure Corinna AI fulfills its core mandate—delivering a secure, intelligent, and highly efficient AI Sales Representative platform at a disruptive price point for the SMB market.

2.3. Current System

The "current system" for small to medium businesses (SMBs) is a combination of fragmented manual processes and existing commercial solutions, each possessing significant weaknesses. **Corinna AI** is positioned to disrupt this status quo by directly addressing the financial, functional, and technical shortcomings of the competition, as summarized below.

Table 2.1 Comparison of Existing Platforms vs. Corinna Ai Solution

Application Name	**Features**	**Weakness**	**Proposed Solution (Corinna AI)**
HubSpot	Create chatbot sequences without any coding. Qualify leads and trigger email campaigns after chatbot interactions.	The **Professional Customer Platform** costs a,whopping $\$1,170$/month, catering to the needs of established businesses, making it financially inaccessible for SMBs.	We will overcome this problem with a **cheaper pricing plan** that will allow entry-level business owners to use similar features at much lower prices.
BotPress	Create chatbots with a drag and	Botpress lacks capabilities like email marketing, **live**	A centralized **Dashboard** that shows

	drop feature or existing templates, creating possible chat-flows.	**chat switch**, and booking meetings, etc.	Pipeline Value, Total sales, metrics, as well as all the missing sales features of Botpress.
Amazon Lex	Create virtual agents and voice assistants chatbots.	It utilizes the AWS cloud platform to deploy the chatbot, hence making it a bit **complex to implement** for non-technical users.	**Corinna AI** offers a single, easy-to-use **code snippet** that, once added to the website, can be easily deployed and used.

Chapter 3

Requirement Analysis

3. Requirement Analysis

To effectively derive the functional requirements for Corinna AI, we will use the Use Case technique. This approach is well-suited for interactive end-user applications, allowing us to define system interactions from the user's perspective.

3.1. Use Case Diagram

A Use Case Diagram will be created using Draw.io to visually represent user interactions with Corinna AI. This will help identify key functionalities, such as chatbot interactions, lead management, email automation, and payment processing.
Here we create a visual representation of the use cases and their relationships using a use case diagram. The Use Case Diagram visualizes the functional requirements of the Corinna AI system, delineating the distinct interactions between the system and its two primary actors: the **Admin** (system administrator) and the **User** (end client/visitor).

- **Admin Functionalities (Backend Management):** The Admin acts as the central controller, possessing comprehensive access to backend management modules. Their primary use cases include **"Login"** for secure access, followed by operational tasks such as **"Manage Users & Authentication"** and **"Manage Domain Settings"** to configure the platform environment. The Admin is also responsible for business oversight, including **"View Dashboard & Business Metrics"** to track performance, **"Manage Pricing Plans"** for monetization, and **"Manage Email Marketing Campaigns"** for outreach. Furthermore, they handle direct client interactions through **"Manage Appointments"** and **"Manage Customer Conversations."**

- **User Functionalities (Frontend Interaction):** The User represents the external visitor interacting with the AI interfaces. Their journey centers on engagement, starting with **"Converse with Chatbot"** to receive immediate assistance. Users can perform transactional actions such as **"Buy Products through Chatbot"** and **"Invest Directly Using Stripe."** Additionally, the system facilitates lead capture and scheduling through the use cases **"Provide Info for Lead Generation"** and **"Book Appointments,"** creating a seamless self-service experience [18].

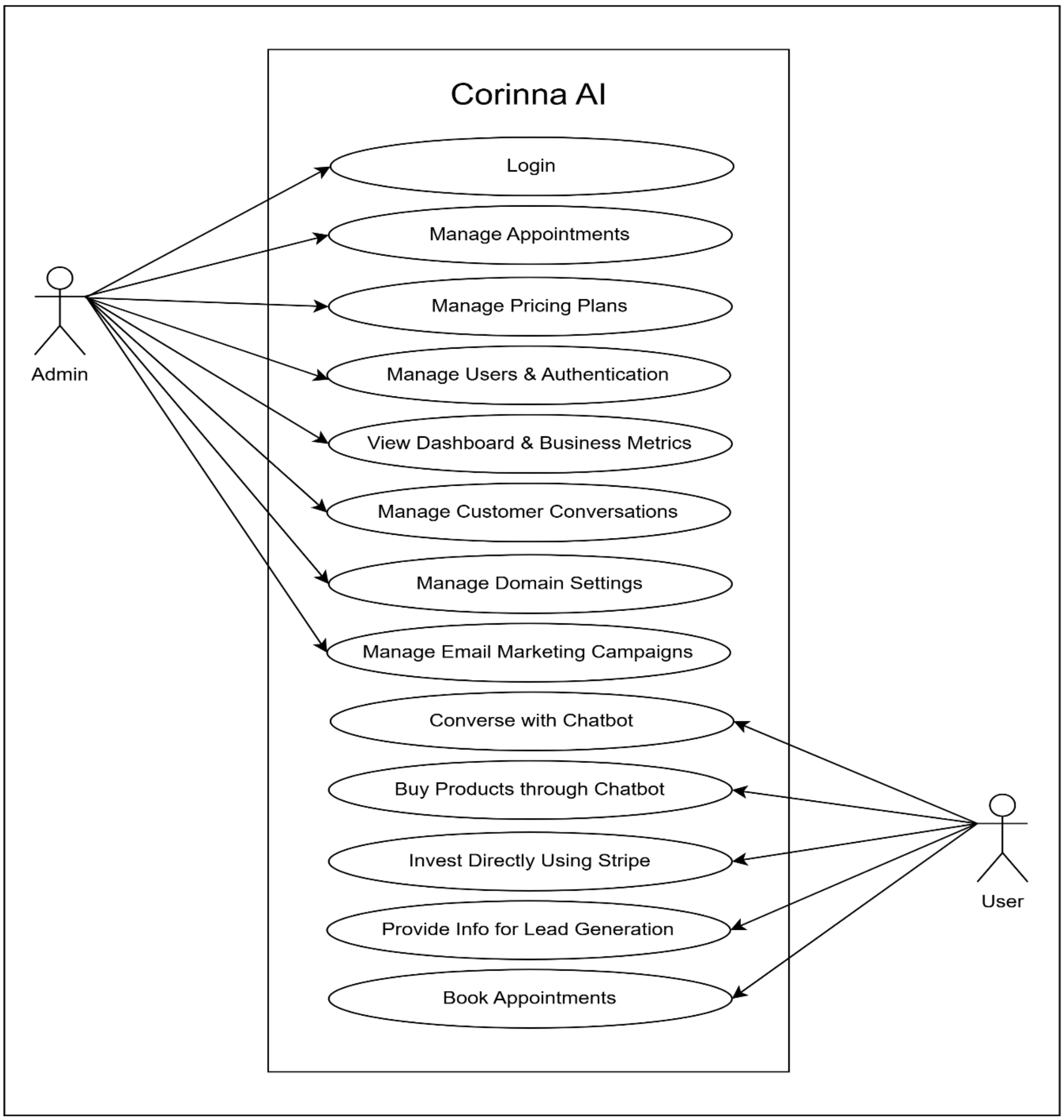

Figure 3.1 Use Case Diagram of Corinna Ai

3.2. Use Case Description

Table 3.1 Login

Use case id	01
Use case name	Login
Actors	- Admin
Description	The admin logs into the system to access Corinna AI features.
Trigger	The admin requests to log in.
Pre-condition	The admin must have an account.
Postcondition	The admin is successfully logged in.
Normal flow	1. The admin enters their email/phone and password. 2. The system validates the credentials. 3. The admin is granted access to the system.
Alternative flow	If credentials are incorrect, an error message is displayed.
Exceptions	If the admin forgets their password, a reset option is provided.
Business rules	The system must be connected to the network.

Table 3.2 Manage Appointments

Use case id	02
Use case name	Manage Meetings
Actors	- Admin
Description	The admin views and manages scheduled meetings booked by users.

Trigger	The admin accesses the appointments tab.
Pre-condition	Users must have booked meetings through the chatbot.
Postcondition	The admin successfully manages appointments.
Normal flow	1. The admin views scheduled meetings. 2. The admin can reschedule or cancel meetings. 3. Notifications are sent to users about any changes.
Alternative flow	If no meetings are scheduled, the admin sees an empty dashboard.
Exceptions	If the system fails to load meetings, an error message is displayed.
Business rules	Meeting changes must be synchronized with the admin's calendar.

Table 3.3 View Dashboard & Business Metrics

Use case id	04
Use case name	View Dashboard
Actors	- Admin
Description	The admin accesses business insights and key metrics.
Trigger	The admin opens the dashboard.
Pre-condition	The system must have collected data.
Postcondition	The admin views business performance metrics.

Normal flow	1. The admin logs into the dashboard. 2. Business metrics are displayed (e.g., leads, sales, transactions).
Alternative flow	If no data is available, an empty dashboard is shown.
Exceptions	If the dashboard fails to load, an error message appears.
Business rules	Business data must be updated in real-time.

Table 3.4 Manage Customer Conversations

Use case id	05
Use case name	Manage Customer Conversations
Actors	- Admin
Description	The admin monitors and takes over chatbot conversations when necessary.
Trigger	The chatbot detects an issue, or the admin manually intervenes.
Pre-condition	The chatbot must be active and receiving user messages.
Postcondition	The admin resolves customer queries manually.
Normal flow	1. The admin accesses the conversation panel. 2. The admin takes over an ongoing chatbot conversation. 3. The admin provides direct responses to the user.
Alternative flow	If the chatbot resolves the issue, no admin intervention is needed.

Exceptions	If the system fails to connect, the admin cannot take over the chat.
Business rules	The admin must respect customer privacy and data security.

Table 3.5 Manage Domains Settings

Use case id	06
Use case name	Manage Website Domains
Actors	- Admin
Description	The admin configures chatbot deployment on various domains.
Trigger	The admin accesses the domain management panel.
Pre-condition	The chatbot must support multiple domains.
Postcondition	The chatbot is successfully linked to a website.
Normal flow	1. The admin adds a new domain. 2. The system generates an embed code. 3. The admin integrates the chatbot into their website.
Alternative flow	If the admin cancels, no domain is added.
Exceptions	If the domain is invalid, an error is displayed.
Business rules	The chatbot must function correctly across all added domains.

Table 3.6 Configure Email Marketing Campaigns

Use case id	07
Use case name	Configure Email Marketing

Actors	- Admin
Description	The admin sets up email campaigns for lead nurturing.
Trigger	The admin accesses the email marketing settings.
Pre-condition	The system must have stored user emails.
Postcondition	The campaign is scheduled or sent.
Normal flow	1. The admin creates an email campaign. 2. The admin selects recipients. 3. The email is sent or scheduled.
Alternative flow	If the admin cancels, no emails are sent.
Exceptions	If email sending fails, an error message appears.
Business rules	Email campaigns must comply with marketing regulations (e.g., GDPR).

Table 3.7 Configure Bot Training

Use case id	08
Use Case Name	Configure Bot Training
Actors	Admin
Description	The admin updates and improves the chatbot’s knowledge base by adding, modifying, or deleting responses.
Trigger	Admin navigates to the bot training section in the settings page
Pre-condition	Admin is logged in and has the necessary permissions.

Post-condition	The chatbot is updated with the latest training data.
Normal Flow	1. Admin adds, modifies, or deletes a question-answer pair. 2. Admin saves the changes. 3. The chatbot updates its knowledge base automatically.
Alternative Flow	If the chatbot training update fails (e.g., due to invalid input), an error message is displayed, and the admin is prompted to retry.
Exceptions	- No internet connection prevents saving updates. - Admin lacks permission to modify chatbot responses.
Business Rules	- The chatbot must always have a default fallback response. - Changes should be version-controlled to prevent accidental data loss.

Table 3.8 Converse with Chatbot

Use case id	09
Use case name	Chat with Chatbot
Actors	- User
Description	The user interacts with the AI chatbot to ask questions, receive responses, or get assistance.
Trigger	The user starts a conversation with the chatbot.

Pre-condition	The chatbot must be active and accessible.
Postcondition	The chatbot provides the required response to the user.
Normal flow	1. The user sends a message to the chatbot. 2. The chatbot processes the input and generates a response. 3. The response is displayed to the user.
Alternative flow	If the chatbot does not understand, it requests clarification.
Exceptions	If the chatbot fails to respond due to system issues, an error message is displayed.
Business rules	The chatbot must be trained with relevant responses.

Table 3.9 Buy Products

Use case id	10
Use case name	Buy Products via Chatbot
Actors	- User
Description	The user purchases products directly through the chatbot interface.
Trigger	The user requests to buy a product.
Pre-condition	The chatbot must have access to the product catalog.
Postcondition	The user completes the purchase, and the payment is processed.
Normal flow	1. The user selects a product through the chatbot. 2. The chatbot provides product details and

	pricing. 3. The user confirms the purchase. 4. The chatbot redirects the user to Stripe for payment. 5. Payment is processed, and confirmation is sent to the user.
Alternative flow	If the user cancels the purchase, the process ends.
Exceptions	Payment failure results in an error message.
Business rules	The product must be available in stock.

Table 3.10 Invest Using Stripe

Use case id	11
Use case name	Invest via Stripe
Actors	- User
Description	The user invests directly using Stripe payment integration.
Trigger	The user is a potential lead detected by the chatbot and clicks on a link
Pre-condition	The user must have a valid payment method.
Postcondition	The payment is successfully processed, and the investment is recorded.
Normal flow	1. The user selects an investment option. 2. The chatbot provides investment details. 3. The user confirms the amount to invest. 4. The chatbot redirects the user to Stripe for payment.

	5. Payment is processed, and the transaction is confirmed.
Alternative flow	If the user cancels, the investment process stops.
Exceptions	If payment fails, an error message is displayed.
Business rules	The payment system must comply with financial regulations.

Table 3.11 Providing Info for future opportunities

Use case id	12
Use case name	Provide Email for Lead Generation
Actors	- User
Description	The user provides their email to the chatbot for marketing and lead generation.
Trigger	The chatbot requests the user's email.
Pre-condition	The user must consent to share their email.
Postcondition	The email is stored for future marketing campaigns.
Normal flow	1. The chatbot asks the user for their email. 2. The user enters their email. 3. The chatbot verifies and stores the email.
Alternative flow	If the user refuses to provide an email, the process ends.
Exceptions	If the email format is invalid, the chatbot requests re-entry.
Business rules	Emails must be stored securely following data protection laws.

Table 3.12 Book Appointments

Use case id	13
Use case name	Book Appointment via Chatbot
Actors	- User
Description	The user books an appointment or meeting through the chatbot.
Trigger	The user requests to schedule a meeting.
Pre-condition	The business owner must have available time slots.
Postcondition	The appointment is successfully scheduled.
Normal flow	1. The user asks to book a meeting. 2. The chatbot displays available time slots. 3. The user selects a suitable slot. 4. The chatbot confirms the booking.
Alternative flow	If no slots are available, the chatbot suggests alternative dates.
Exceptions	If the booking fails, an error message is displayed.
Business rules	Appointments must be synchronized with the business owner's calendar.

3.3. Functional Requirements

Corinna AI is designed as a modular, AI-powered platform that enhances customer engagement, lead management, and sales automation. The system must meet the following functional requirements:

1. Meeting Booking

- Users can schedule appointments via the AI chatbot.
- The system must sync with the business owner's calendar.
- Users receive confirmation and reminders for booked meetings.

2. Pricing Plans

- The system must offer tiered subscription plans.
- Users can view feature details and upgrade their plans.

3. Authentication & Authorization

- Supports user registration, login, and secure password management.
- Implements role-based access control for different user levels.

4. Dashboard

- Displays key business metrics, such as potential clients, sales, and transactions.
- Users can track subscription usage and pipeline value.

5. Conversations

- Allows business owners to manage real-time customer interactions.
- Chatbot must detect dissatisfaction and escalate to a human agent when necessary.

6. Domains Management

- Enables users to add and manage website domains.
- Provides a code snippet for chatbot integration.
- Users can train the chatbot with FAQs and product information.

7. Integrations

- Connects with third-party services like **Stripe** for payment processing.

8. Settings

- Users can update billing details and change subscription plans.
- Allows interface customization, including dark mode.

9. Appointments

- Displays upcoming and scheduled meetings in a dedicated interface.

10. Email Marketing

- Captures lead through chatbot interactions.
- Allows users to create and track email campaigns.

11. Product Selling / In-Chat Sales Automation

- The chatbot can showcase products or services to customers.
- Users can view product details, pricing, and availability.
- The chatbot can guide customers through the purchase process.
- Customers are redirected to a secure checkout page when ready to buy.
- The system records product-related inquiries for sales insights.

3.4. Non-Functional Requirements

Non-functional requirements (NFRs) define the quality attributes, system constraints, and operational capabilities of Corinna AI. Below are the key NFRs categorized appropriately:

1. Performance Requirements

- The chatbot should respond to user queries within 2 seconds for a seamless experience.
- The dashboard should be loaded within 3 seconds under normal usage conditions.
- The system must support at least 1000 concurrent users without performance degradation.

2. Security Requirements

- All user data, including passwords and payment details, must be encrypted using AES-256.
- The system must enforce role-based access control (RBAC) to restrict unauthorized access.
- Users and admins must be automatically logged out after 30 minutes of inactivity.
- Payment transactions must comply with PCI DSS (Payment Card Industry Data Security Standard).

3.Availability & Reliability Requirements

- The system should have 99.9% uptime, ensuring minimal downtime.
- Corinna AI should be accessible 24/7, with scheduled maintenance only during low-traffic hours.
- In case of failure, the system should recover within 5 minutes using automated backup mechanisms.

4. Scalability Requirements

- The platform should be able to **scale horizontally** to handle increased traffic.
- The AI chatbot should be capable of processing at least **10,000 messages per hour**.
- The email marketing system should support sending at least **500,000 emails per day** without failure.

5. Usability Requirements

- The user interface must be **intuitive and easy to navigate** with a maximum of **3 clicks to reach key features**.
- The chatbot should support **natural language processing (NLP)** to improve user interaction.
- The system must provide a **dark mode and light mode** option for accessibility.

Chapter 4

Design and Architecture

4. Design and Architecture

In this section, we explore design models that provide a comprehensive understanding of the architecture and interactions of the **Corinna AI** project. These models offer visual representations of the system's structure and behavior, facilitating comprehension of its complex components and processes [19].

4.1. System Architectural Design

The **Corinna AI** platform utilizes a **Modular Monolithic Architecture** built on a **Client-Server** model. This approach allows for the unified development of the frontend and backend within a single **Next.js** framework while maintaining strict separation between business logic modules. This design was chosen to balance the ease of development and deployment with the scalability required for a SaaS product [20].

The system is structurally divided into four primary layers, each with specific responsibilities:

1. Client Layer (Presentation)

This layer represents the user-facing interface accessible via any modern web browser. It is built using **React** components and styled with **Tailwind CSS**.

- **Agent Dashboard:** The administrative interface for business owners to manage leads, view analytics, and configure the chatbot.
- **Chat Widget:** The lightweight, embeddable component that runs on client websites to interact with visitors.
- **Responsibilities:** Rendering the UI, handling user input events, and communicating with the server via secure **HTTPS** requests and **WebSockets** (for real-time updates).

2. Application Layer (Business Logic)

This is the core of the application, hosted as serverless functions on **Vercel**. It processes all incoming requests from the client layer.

- **API Gateway (Next.js API Routes):** The entry point for all backend logic.
- **Authentication Middleware (Clerk):** Intercepts requests to validate user sessions and enforce Role-Based Access Control (RBAC) before they reach the business logic.
- **Business Modules:** The logic is segmented into distinct engines to ensure separation of concerns:
  - **Chatbot Engine:** Manages conversation flow and AI interaction.
  - **Scheduling Engine:** Handles calendar syncing and appointment creation.
  - **Sales Engine:** Processes lead qualification and scoring.
  - **Payment Controller:** Manages subscription status and billing cycles.

3. Data Layer (Persistence)

This layer is responsible for the reliable storage and retrieval of application data.

- **Neon (PostgreSQL):** The primary relational database used to store structured data such as user profiles, lead records, conversation history, and appointments [21].
- **Prisma ORM:** Acts as the abstraction layer between the Application and Data layers, providing type-safe database queries and schema management.
- **Redis (Optional/Planned):** Utilized for high-speed caching of active chat session states to reduce database load during high-traffic periods [22].

4. External Services Layer

To provide enterprise-grade functionality without reinventing the wheel, the architecture tightly integrates specialized third-party APIs:

- **OpenAI API:** Provides the Large Language Model (GPT-3.5/4) for generative text responses.
- **Stripe:** Handles all PCI-compliant payment processing and subscription management.
- **Pusher:** Facilitates the real-time, bi-directional event streaming required for the **Live Chat Handoff** feature [23].
- **Uploadcare:** Manages the secure storage and delivery of large files, such as PDF documents used for chatbot training.
- **Clerk:** Manages secure user identity, session tokens, and authentication flows.

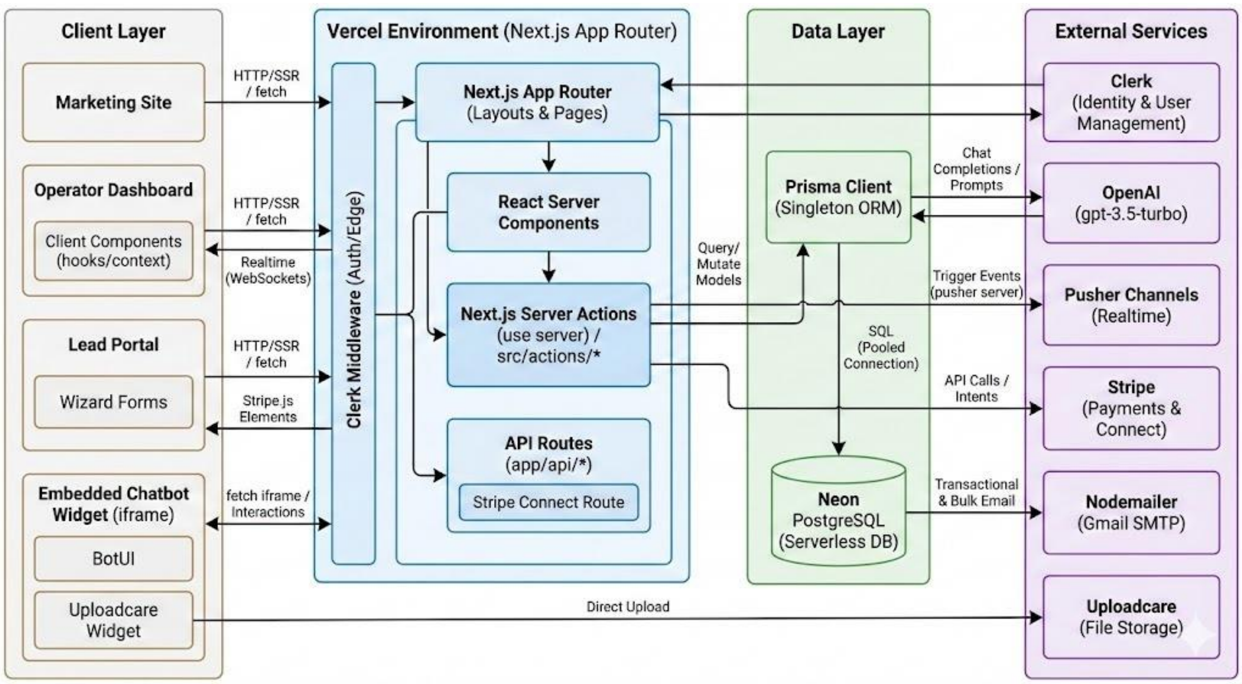

Figure 4.1 System Architecture Design

4.2. Data Representation

The data model for Corinna AI is designed to support a multi-tenant SaaS architecture, ensuring strict data isolation between different business clients while maintaining relational integrity for complex sales data.

The core entities and their relationships are defined as follows:

- **Organization (Tenant Root):** The top-level entity representing a business client. It holds global settings, subscription status (Free/Pro/Ultimate), and is the parent record for all other data. It has a **One-to-One** relationship with the **Subscription** table (managed via Stripe IDs).
- **Chatbot:** Each organization has one or more chatbots. This entity stores the configuration data, including the welcomeMessage, icon, background color, and references to the training data. It shares a **One-to-Many** relationship with **Conversations**.
- **Conversation:** Represents a unique chat session initiated by a website visitor. It tracks the current state (Active, Completed) and the critical live boolean flag, which determines if the chat is in AI mode or Real-Time Human mode.
- **Message:** Individual text entries linked to a Conversation. Each message stores the content, the role (user vs. assistant), and a timestamp, creating a complete audit trail of the interaction.
- **Lead:** When a visitor provides contact information, a Lead record is created and linked to the Conversation. This entity stores the email, name, and current status in the sales funnel.
- **Appointment:** Stores scheduling details (date, time) booked through the chat, linked directly to a specific Customer/Lead.

4.2.1. **Level 0 DFD**

This Data Flow Diagram (DFD) Level 0, or Context Diagram, provides a high-level overview of the Corinna AI Platform and its interactions with external entities.

At the center is the **Corinna AI Platform**, represented as a single process (node 0), signifying the entire system. Surrounding this central process are all the external entities that either provide data to or receive data from the Corinna AI Platform.

Here's a breakdown of the interactions:

- **Site Visitor (Prospect)**: This entity represents the end-users interacting with the AI chatbot on a client's website. They send "Chat Messages," "File Uploads," and "Form Data (Bookings)" to the Corinna AI Platform. In return, the platform sends "AI Responses," "Appointment Confirmations," and "Payment Links" back to the Site Visitor.
- **Domain Owner (Operator)**: This entity represents the business users who manage the chatbot and leads. They provide "Configuration Settings" (e.g., chatbot tone, filter questions), "Live Chat Replies," and "Campaign Requests" to the platform. The Corinna AI Platform, in turn, provides "Lead Dashboards," "Real-time Alerts," and "Campaign Analytics" to the Domain Owner.
- **OpenAI Service**: This external service is an AI provider. The Corinna AI Platform sends "Prompts & Chat History" to OpenAI, and OpenAI returns "AI Completions (Text Responses)" to the platform.
- **Stripe**: This is an external payment gateway. The Corinna AI Platform sends "Payment Intents" and "Account Connect Requests" to Stripe. Stripe then returns "Transaction Status" and "Payout Data" to the platform.
- **Pusher**: This is an external real-time WebSocket service. The Corinna AI Platform sends "Event Triggers" (e.g., "New Message," "Realtime Mode") to Pusher, indicating real-time events within the system.
- **Email Service (Gmail/SMTP)**: This represents an external mail provider. The Corinna AI Platform sends "Email Payloads (Campaigns, Alerts)" to the Email Service for delivery.
- **Clerk**: This is an external authentication provider. The Corinna AI Platform exchanges "Auth Tokens" and "User Session Data (Verification)" with Clerk to manage user identities and sessions.

In essence, the diagram illustrates the primary inputs, outputs, and external systems that the Corinna AI Platform interacts with to perform its core functions.

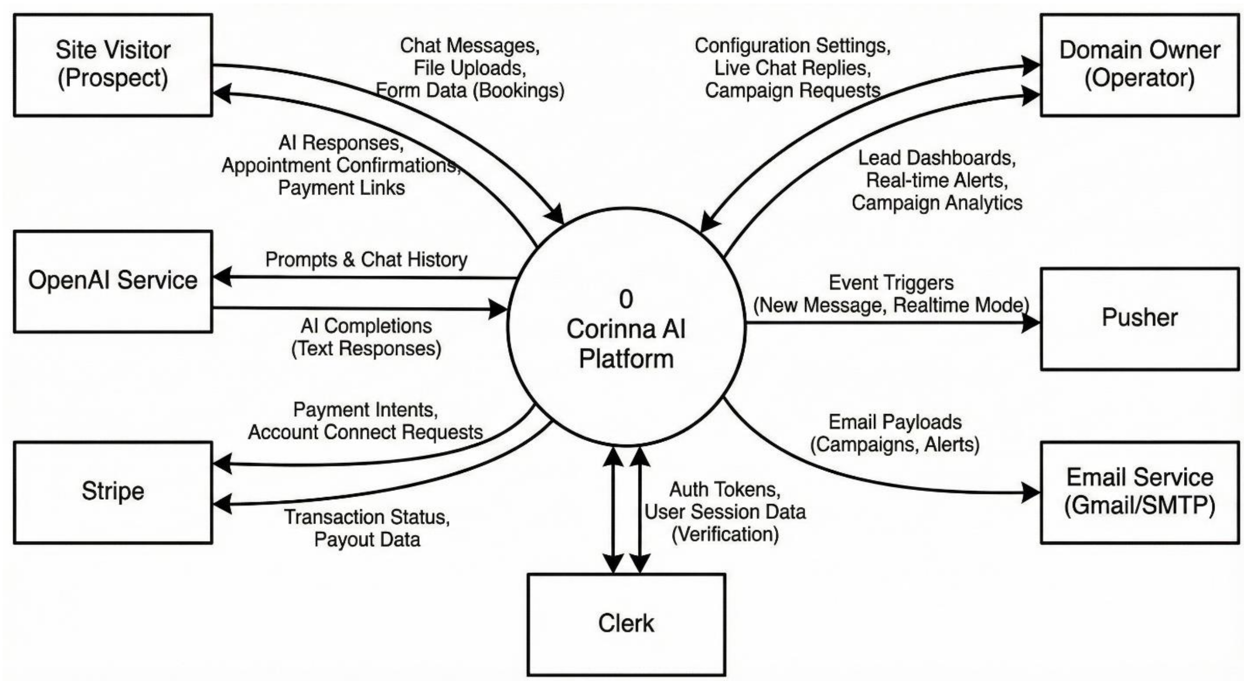

Figure 4.2 Level 0 DFD

4.2.2. **Level 1 DFD**

This Data Flow Diagram (DFD) Level 1 provides a more detailed breakdown of the system's core functionalities, expanding upon the high-level overview of a Level 0 DFD. It illustrates how five key processes interact with each other, external entities, and a central database.

Central Database (D1 - Central Database: Neon/Postgres): This is the heart of the system, serving as the primary repository for critical data including Users, Domains, ChatRooms, Customers, and Bookings. All processes interact with this database to read or write information.

Processes:

- **1.0 Manage Authentication:** This process is responsible for user login and session validation. It receives credentials from external Users/Owners (often facilitated by a Clerk service). Upon successful authentication, it outputs Validated Session Tokens that grant access to other system processes. It also writes New User Accounts to the D1 Central Database.

- **2.0 Process Chat & AI Workflow:** This is the core chatbot engine. It receives Visitor Messages from the Widget/Visitors. To provide relevant responses, it queries D1 for Domain Settings & Filter Questions. It then sends Prompts to an external OpenAI service and receives Answers back. This process is crucial for capturing Chat Logs & New Leads (Customers), which are written to D1. It also triggers 5.0 Handle Communications for real-time updates.
- **3.0 Manage Dashboard & Settings:** This process represents the administrative interface for operators or owners. It receives Domain Configs & Marketing Templates from the Owner. It reads Lead Data and Chat History Query from D1 to display on the dashboard, providing valuable insights. Additionally, it updates Billings and Product Catalogs in D1.
- **4.0 Process Portal Transactions:** This process handles public-facing interactions such as appointments and payments. It receives Booking Requests & Payment Details from Visitors. Before confirming a booking, it checks Availability Slots Query from D1. It interacts with Stripe for Payment Processing Requests and receives Payment Confirmations. Finally, it writes Confirmed Bookings to D1.
- **5.0 Handle Communications:** This process manages outbound communications. It receives Trigger Events from other processes (e.g., from Process 2.0 or 3.0). It sends Broadcast Events to an external Pusher service for real-time notifications. It also reads Campaign Templates Query from D1 and uses an Email Service to send out Emails.

This DFD Level 1 provides a clear visual representation of the system's operational components and their interdependencies, showing how data flows through the various stages of user interaction, internal processing, and external service integration.

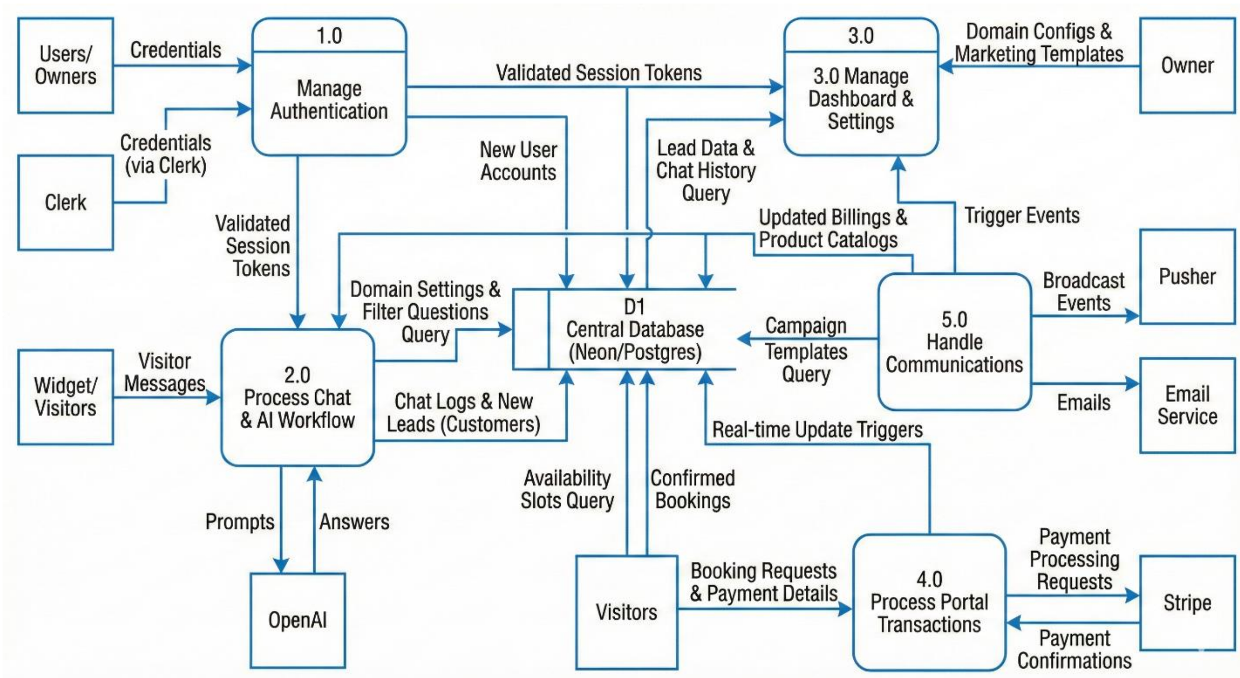

Figure 4.3 Level 1 DFD

4.2.3. **Level 2 DFD**

Drill-down of Process 2.0: Process Chat & AI Workflow

This diagram details the lifecycle of a message entering the chat widget, from initial filtering to final persistence.

- Visitor Message enters 2.1 Identify & Filter. It checks D1 for an existing Customer record.
- Data flows to 2.2 Context Construction, which pulls FilterQuestions and recent ChatHistory from D1.
- The constructed prompt flows to 2.3 AI Interaction, which sends it to OpenAI and gets a text reply.

The reply flows to **2.4 Response Analysis**:

- If the reply contains (realtime), a flow triggers **Process 5.0** to alert the human agent.
- If the reply contains (complete), a flow updates the **CustomerResponses** table in **D1**.

- Finally, the **2.5 Data Persistence** process saves the **ChatMessage** (both user and assistant) into the **ChatRoom** table in **D1**.

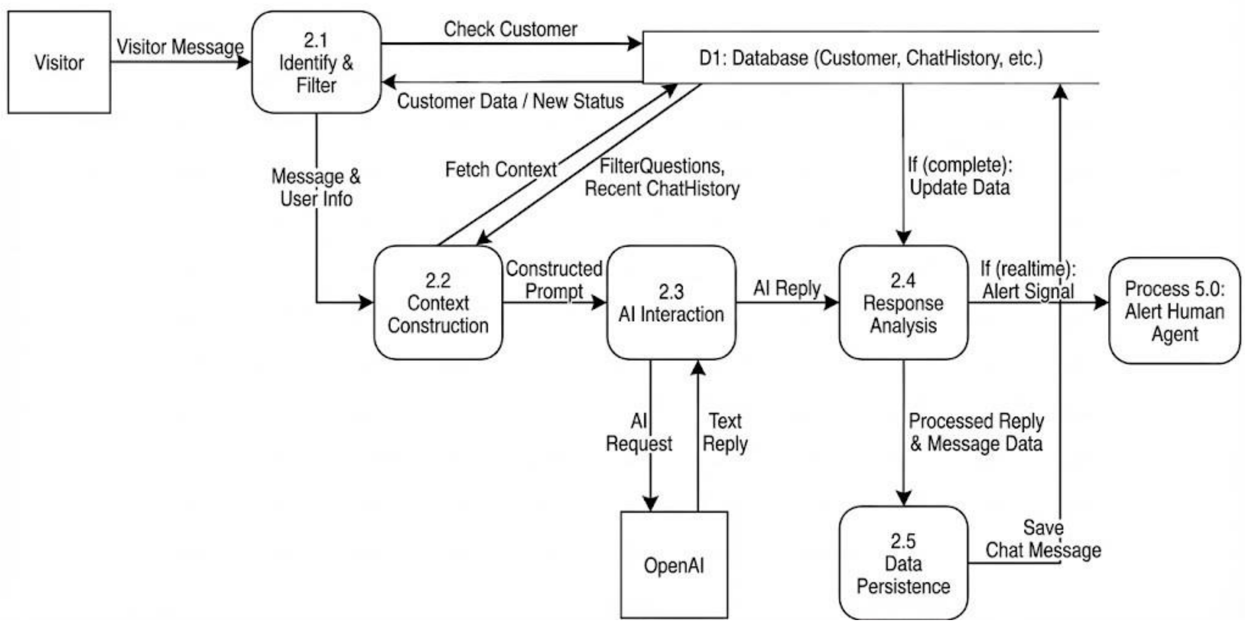

Figure 4.4 DFD Level 2 Process Chat & AI Workflow

Drill-down of Process 3.0: Manage Dashboard & Settings (Marketing & Config)

This diagram focuses on how an owner sets up their domain and runs email campaigns, including the necessary credit checks.

- Owner inputs settings into 3.1 Domain Configuration. This updates Domain, ChatBot, and HelpDesk tables in D1.
- Owner initiates a campaign via 3.3 Campaign Execution.
- Process 3.3 first queries 3.2 Credit Management to check the Billings table in D1 for sufficient credit.
- If approved, 3.3 reads Customer emails from D1, formats the template, and passes the payload to the Email Service.
- Credit Management then decrements the credit count in the Billings table.

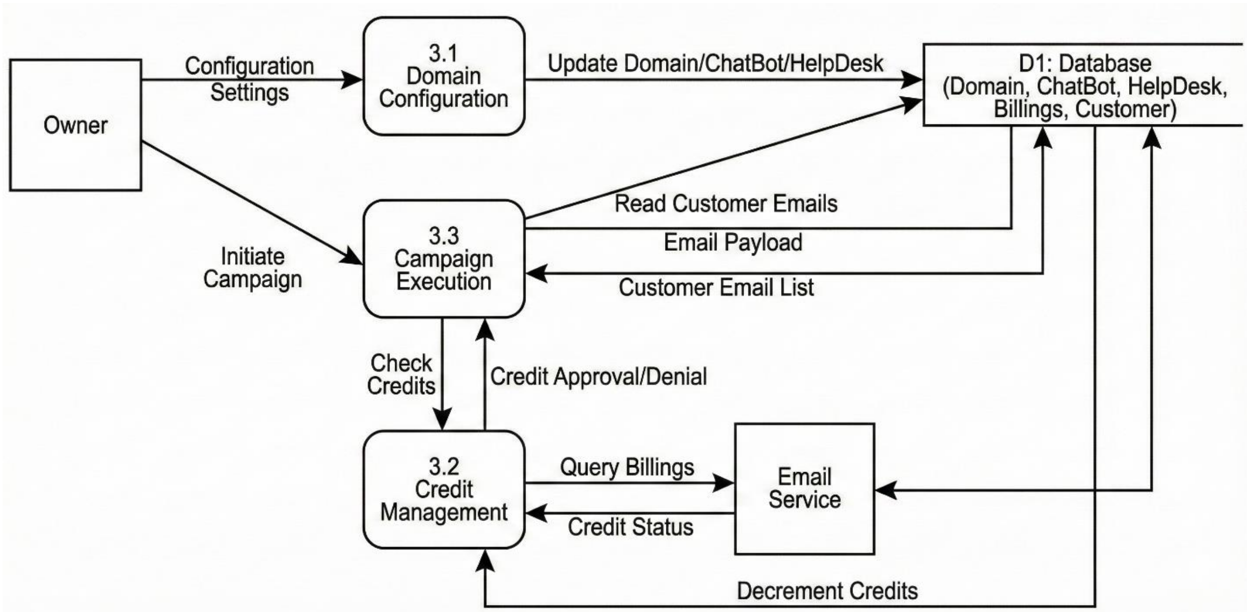

Figure 4.5 DFD Level 2 Manage Dashboard & Settings

Drill-down of Process 4.0: Process Portal Transactions

This diagram illustrates the public-facing "Lead Portal" where appointments and payments happen, from slot verification to payment confirmation.

- **Visitor**s select a time. Data flows to **4.1 Slot Verification**, which queries the **Bookings** table in **D1** to ensure no conflicts exist.
- If free, the flow moves to **4.2 Appointment Booking**, which creates a pending entry in **D1**.
- If the service requires payment, data flows to **4.3 Payment Processing**.
- **4.3** sends a request to **Stripe**. Upon a success signal from **Stripe**, **4.3** updates the **Bookings** table in **D1** to "Confirmed" and links the transaction to the **Product** table.

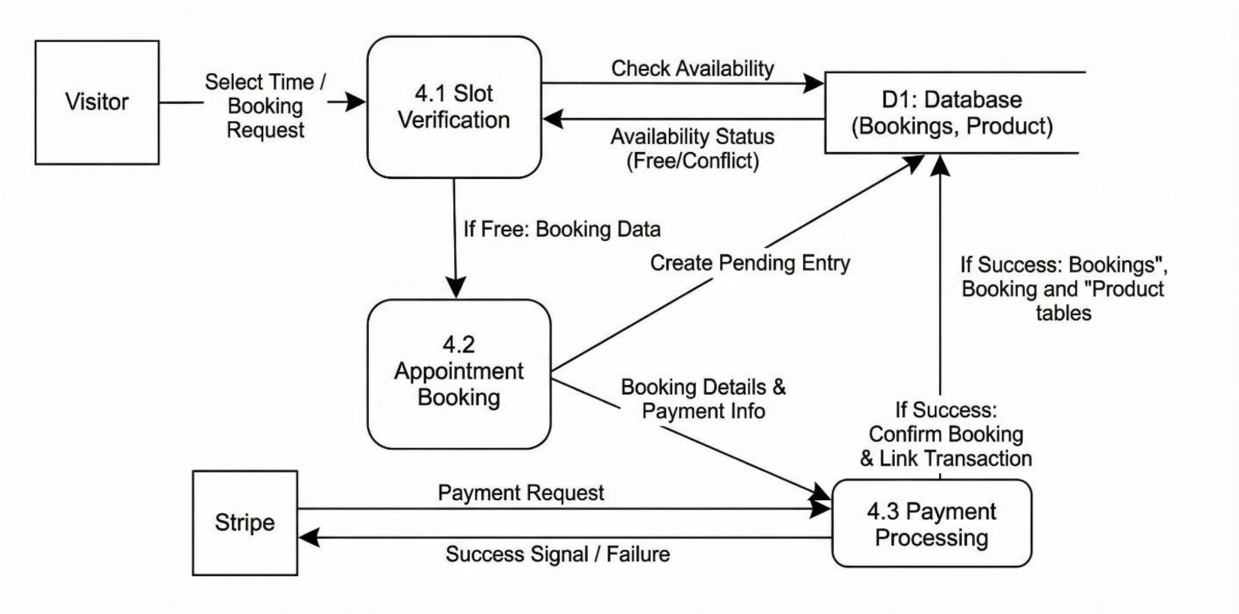

Figure 4.6 DFD Level 2 Process Portal Transactions

4.3. **Process Flow**

A visual representation of the workflow is essential to grasp the complexities of the Corinna AI platform. The activity diagram below depicts the sequential steps and user interactions throughout the system's core processes.

The process flow diagram illustrates the complete journey of a user interacting with the Corinna AI platform from first visiting the website to managing domains, chatbot settings, appointments, email marketing, and real-time conversations. It represents how the system guides users through onboarding, dashboard navigation, and feature-specific modules.

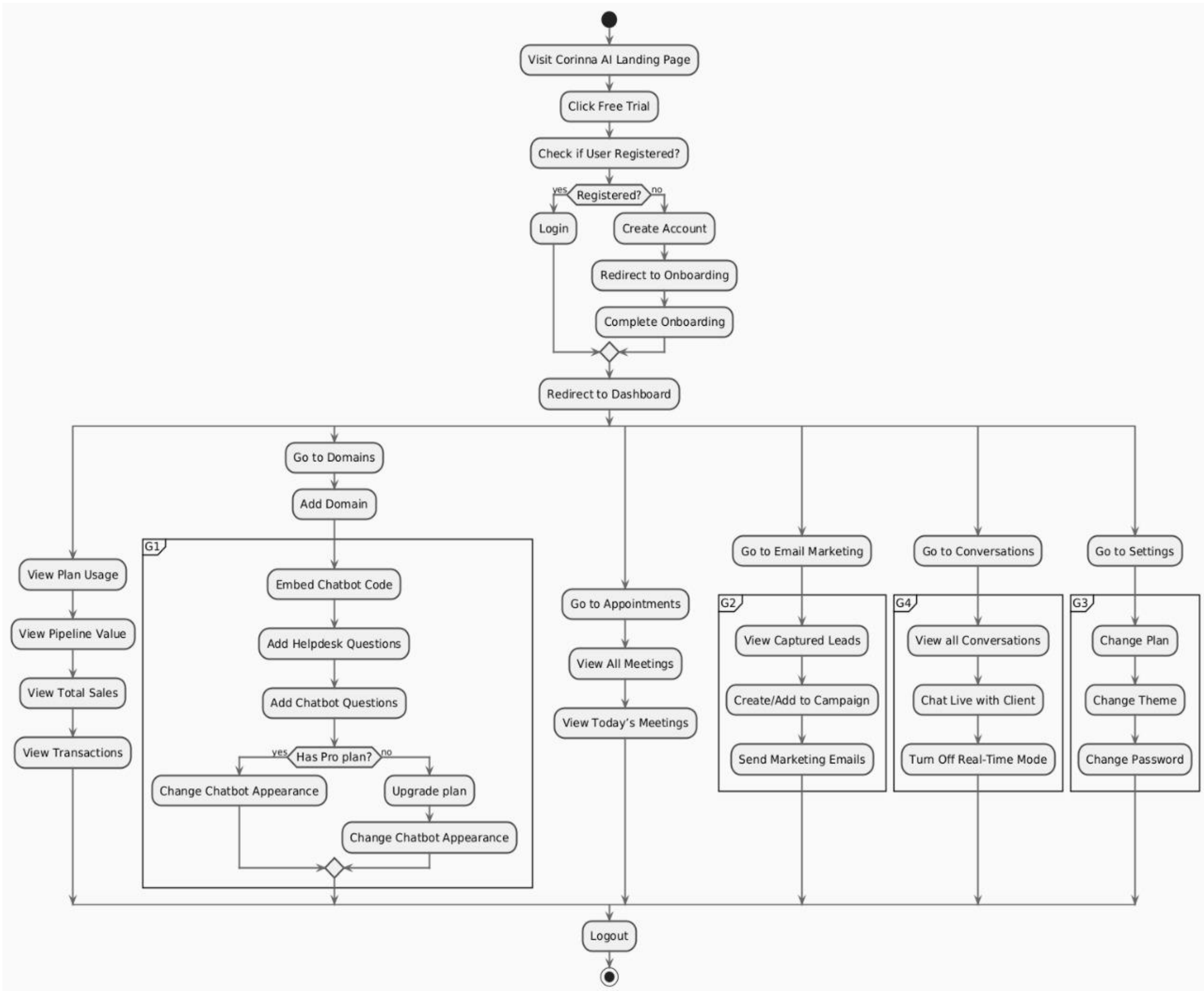

Figure 4.7 Process Flow Diagram

1 Landing Page → Trial & Registration

- The user begins at the **Corinna AI Landing Page**.
- They click **"Free Trial"** to explore the platform.
- The system checks whether the user is already registered:
    1. **If registered** → user logs in.
    2. **If not registered** → user creates a new account.

After account creation, the user is automatically redirected to **Onboarding**.

2 Onboarding Process

The onboarding flow ensures the user sets up the essentials of their Corinna AI workspace:

- Guided onboarding screens collect business details.
- After completing onboarding, the system redirects the user to the **Dashboard**.

This marks the beginning of the user's main workflow.

3 Dashboard Overview

From the dashboard, the user can access all modules of the platform:

- Plan Usage
- Domains
- Appointments
- Email Marketing
- Conversations
- Settings

This structure aligns with your SaaS architecture where each feature is a separate module.

4 Plan Usage & Analytics Module

This left-most flow shows the analytics section:

- **View Plan Usage** → See monthly message limits, domain limits, features available based on subscription.
- **View Pipeline Value** → Total estimated value of captured leads.
- **View Total Sales** → Track sales conversions attributed to the AI.
- **View Transactions** → History of payments/subscriptions.

This module helps SMBs track performance and see ROI of the AI assistant.

5 Domain Management Module

Users can create and manage domains (websites where the chatbot will be installed):

1. **Go to Domains**
2. **Add Domain**

   Inside domain configuration (G1 block), the user can:

- **Embed Chatbot Code** into their website
- **Add Helpdesk Questions** (FAQs)
- **Add Chatbot Questions/Training Data**

Pro Plan Logic

The flow checks whether the user has a Pro plan:

- **If Yes** →
  The user can **change chatbot appearance**, customize theme, bubble colors, widget look, and disable branding.
- **If No** →
  The system prompts to **Upgrade Plan** before allowing appearance changes.

This matches your monetization model (Basic vs Pro).

6 Appointment Scheduling Module

In the appointments section, the user can:

- **View All Meetings** booked through the chatbot
- **View Today's Meetings**

Corinna AI automatically offers meeting booking links during chat, so this module manages all scheduled sessions.

7 **Email Marketing Module**

This part represents your integrated email automation (G2 block):

- **View Captured Leads** → List of emails extracted by the AI chatbot
- **Create/Add to Campaign** → Build email sequences
- **Send Marketing Emails** → Send bulk or automated follow-up emails

This integration eliminates the need for external email tools, giving your platform a competitive edge.

8 **Conversations Module**

The Conversations section (G4 block) manages all chat interactions:

- **View All Conversations** → See AI–customer chat history
- **Chat Live with Client** → Human operator can join the chat
- **Turn Off Real-Time Mode** → Disable live human intervention

This module supports **AI-Human hybrid communication**, a key feature of your system.

9 **Settings Module**

In the settings area, users can:

- **Change Plan** → Upgrade or downgrade subscription
- **Change Theme** → Dashboard UI customization
- **Change Password** → Account security management

10 **Logout**

At any point, the user can safely end their session, which completes the flow.

4.4. Design Models

The following diagrams illustrate the high-level interaction flows between visitors and business owners within the Corinna AI platform, highlighting how various system modules collaborate to deliver core functionalities

4.4.1. Sequence Diagram

This section outlines the end-to-end workflow of the SaaS AI Sales Representative Chatbot platform. The system facilitates interactions between website visitors and business owners, covering account creation, chatbot deployment, automated sales interactions, and lead management.

A. System Actors

Visitor: A potential customer landing on the external website where the chatbot is embedded.

Business Owner: The registered user of the SaaS platform who manages the chatbot, settings, and leads.

B. Functional Modules and Workflows

1. Account Creation and Dashboard Access

This module handles the initial onboarding of a new Business Owner.

- **Trigger:** Visitor clicks "Free Trial" on the landing page.
- **Process:**
    - **Redirect:** The system redirects the user to the Onboarding Page.
    - **Account Creation:** The user completes the sign-up steps.

- **Access:** Upon success, the system redirects the user to the main Dashboard, establishing them as a "Business Owner."

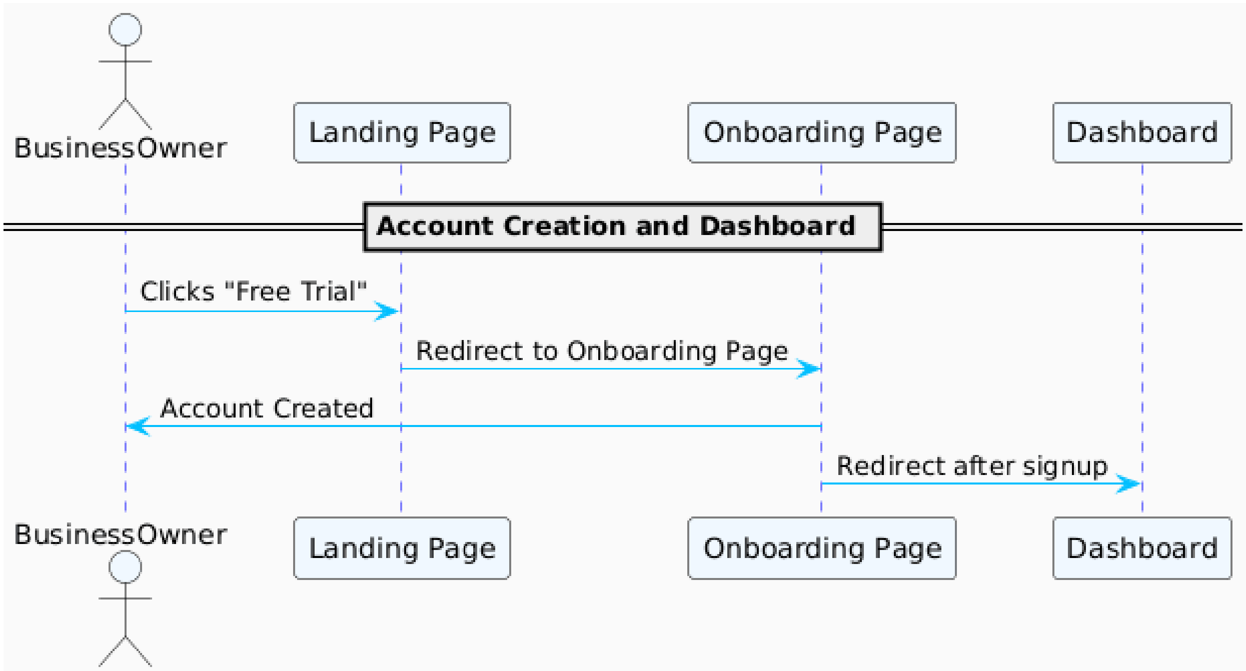

Figure 4.8 Account Creation and Dashboard Access

2. Adding Domain and Embedding Chatbot

This process describes how the chatbot is deployed to an external website.

- **Trigger:** Business Owner clicks "+ Add Domain" on the Dashboard.
- **Process:**
  - **Configuration:** The system opens the Domain Settings page.
  - **Code Generation:** The platform generates and displays a unique JavaScript embed code snippet.
  - **Deployment:** The Business Owner copies and embeds this snippet into their website's HTML.

- **Activation:** The chatbot becomes active and visible on the client's domain.

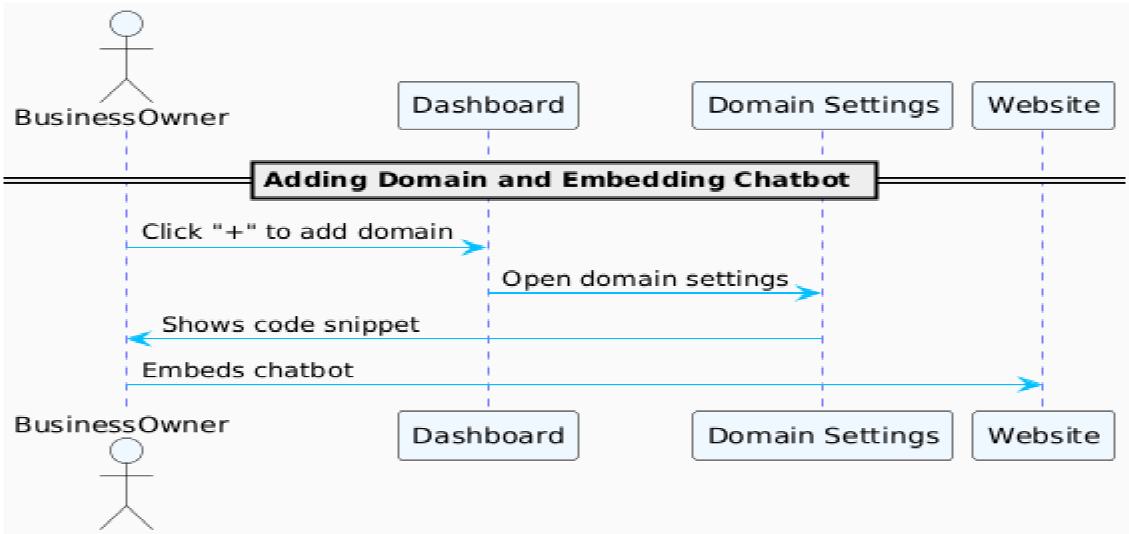

Figure 4.9 Adding Domain and Embedding Chatbot

3. Chatbot Interaction Flow

This is the core operational flow where the AI interacts with a Website Visitor. The system analyses user input to determine intent.

- **Initial Contact:**
    - Visitor sends a message.
    - Chatbot requests contact details (email) and answers initial questions.
    - System performs **Intent Analysis**.
- **Scenario A: Meeting Intent**
    - **Detection:** User expresses interest in booking a call or demo.
    - **Action:** Chatbot provides a calendar booking link.
    - **Outcome:** User books a slot via the Calendar Page; system redirects to a "Thank You" page.
- **Scenario B: Purchase Intent**
    - **Detection:** User expresses readiness to buy.
    - **Action:** Chatbot generates and sends a secure payment link.

- **Outcome:** User completes transaction via Payment Page; system redirects to success page.

- **Scenario C: Escalation (Unhappy Visitor)**
  - **Detection:** Sentiment analysis detects frustration or repeated negative feedback.
  - **Action 1:** System triggers an email notification to the Business Owner.
  - **Action 2:** System enables **Real-Time Mode**.
  - **Outcome:** Business Owner manually joins the conversation via the Conversations module to resolve the issue.

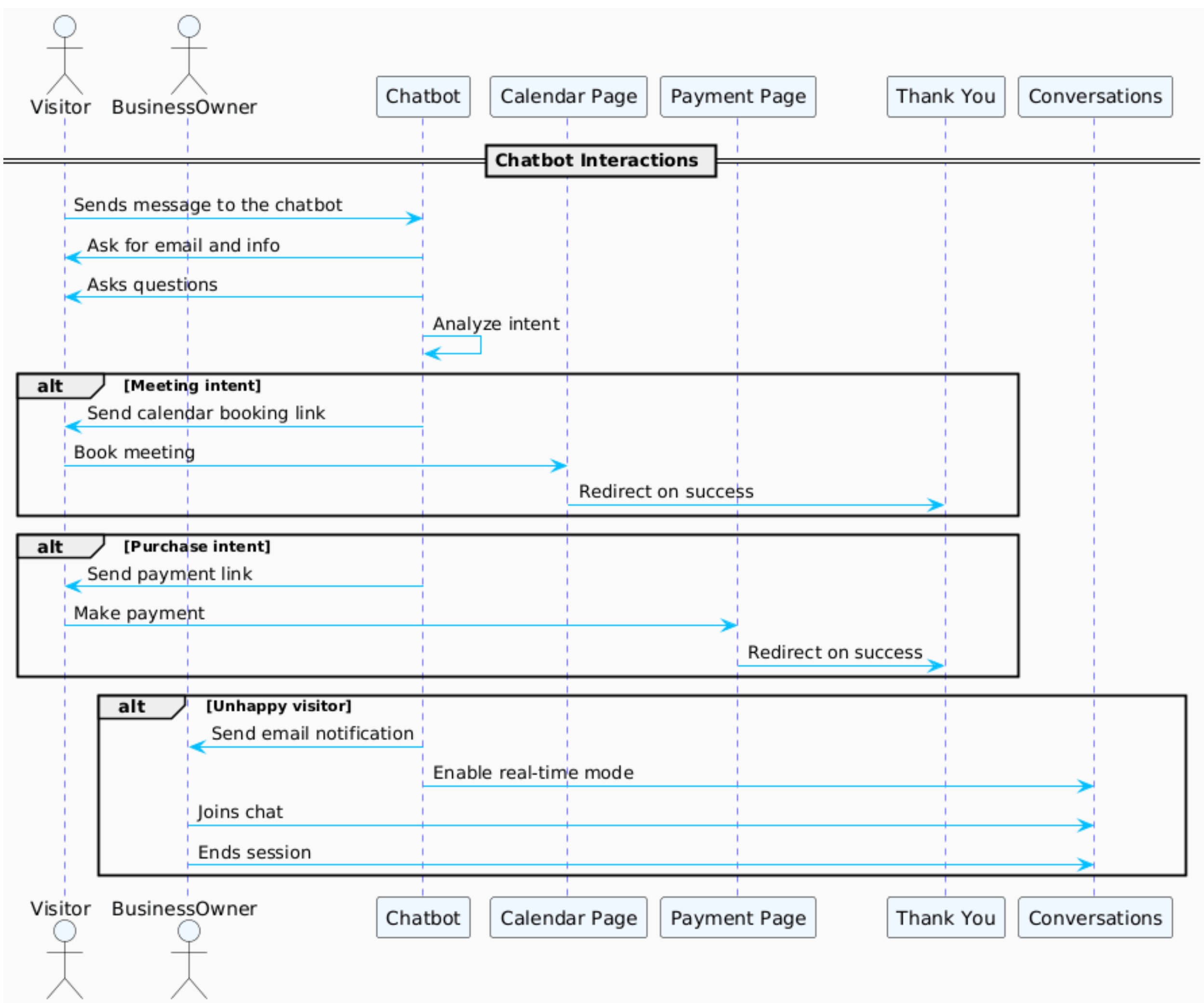

Figure 4.10 Chatbot Interaction Flow

4. Customization for Pro/Ultimate Plans

Allows Business Owners to tailor the chatbot identity and manage billing.

- **Customization Options:**
    - Modify Bot Icon and Greeting Message.
    - Upload FAQs and Canned Responses.
    - Input Product Information for AI context.
- **Stripe Integration:**
    - **Condition:** If payment processing is unlinked.
    - **Process:** Business Owner clicks "Connect Stripe" -> System redirects to Stripe Onboarding -> Account is linked -> Redirect back to Dashboard

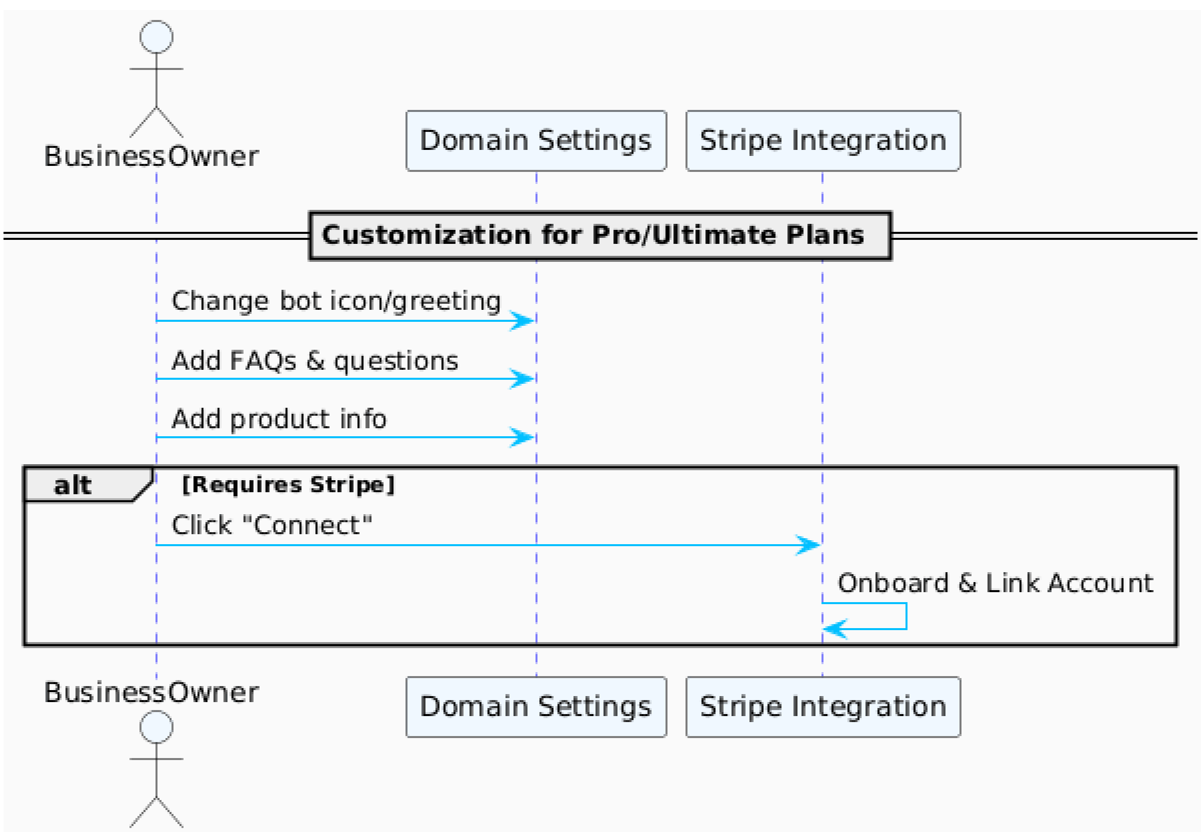

Figure 4.11 Customization for Pro/Ultimate Plans

5. Platform Management Features

General administrative functions for the Business Owner.

- **Subscription:** Change plan (Standard, Pro, Ultimate) via Settings; redirects to Payment Form.
- **Security:** Update account password.
- **Interface:** Toggle Dark/Light mode or Split-Aware mode.

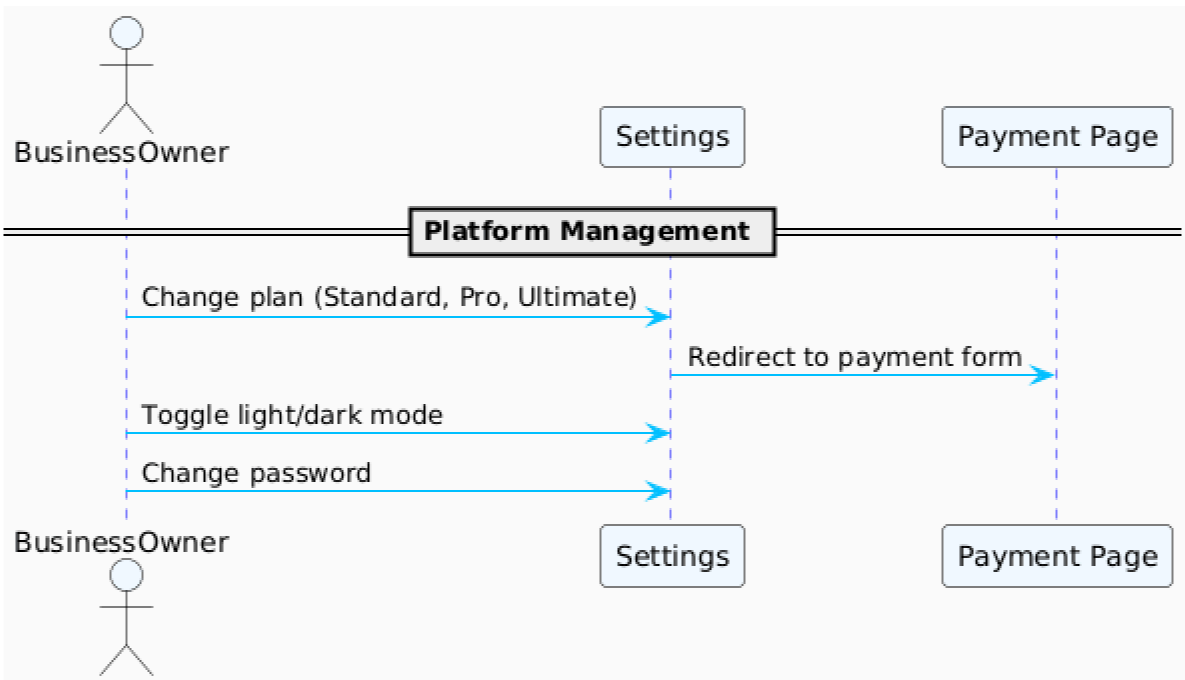

Figure 4.12 Platform Management Features

6. Appointment Management

- **Source:** Bookings generated automatically by the Chatbot or manual entry.
- **Functionality:** Business Owner views daily and upcoming meetings via the **Appointments** module.

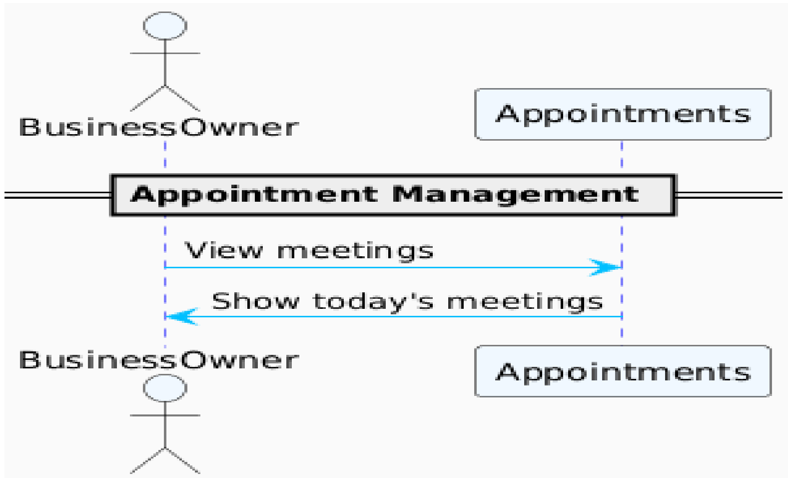

Figure 4.13 Appointment Management

7. **Email Marketing & Lead Management**
   - **Data Collection:** The **Leads Module** automatically captures visitor emails, questions, booking intents, and conversation logs.
   - **Campaign Management:**
     - View captured leads.
     - Add leads to specific marketing campaigns.
     - Create and send automated email sequences to nurture leads.

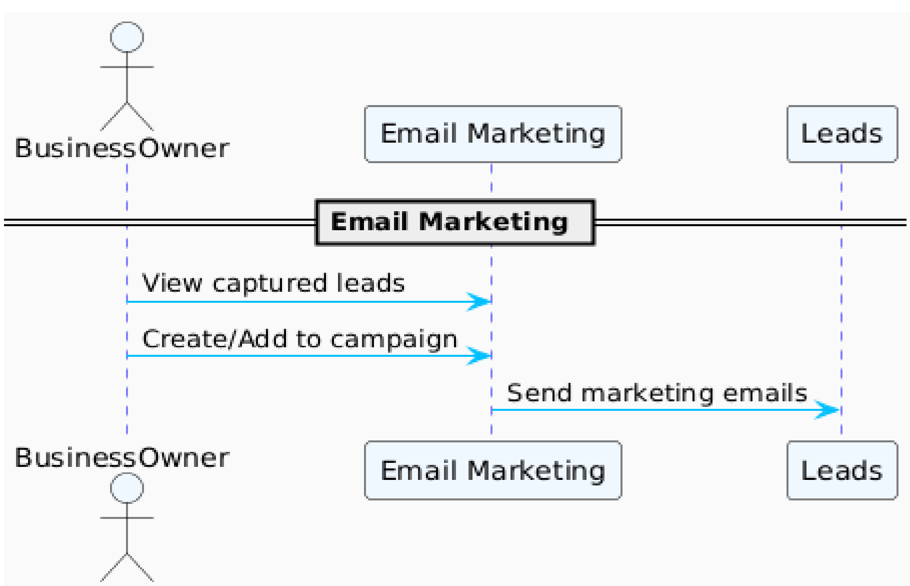

Figure 4.14 Email Marketing & Lead Management

4.4.2. **Class Diagram**

The Class Diagram provides a static view of the system's architecture, illustrating the relationships between the database entities, business logic services, user interface components, and external integrations. The diagram is categorized by color to distinguish between the different layers of the application architecture.

Key Entities:

- **Classes and Models**: Represented with yellow boxes containing fields (attributes) and operations (methods), such as User, Campaign, ChatRoom, Product.
- **Services**: Shown in green (e.g., AuthService, ConversationService, AppointmentService), indicating they provide functionality via operations.
- **Components**: Blue boxes (e.g., AIChatBot, ChatBubble, RealTimeMode), likely UI or functional components.
- **Enums**: Red boxes (e.g., Plans, Role) define enumerated types.
- **External Services**: Grey boxes for integrations (e.g., PusherServer, PusherClient).

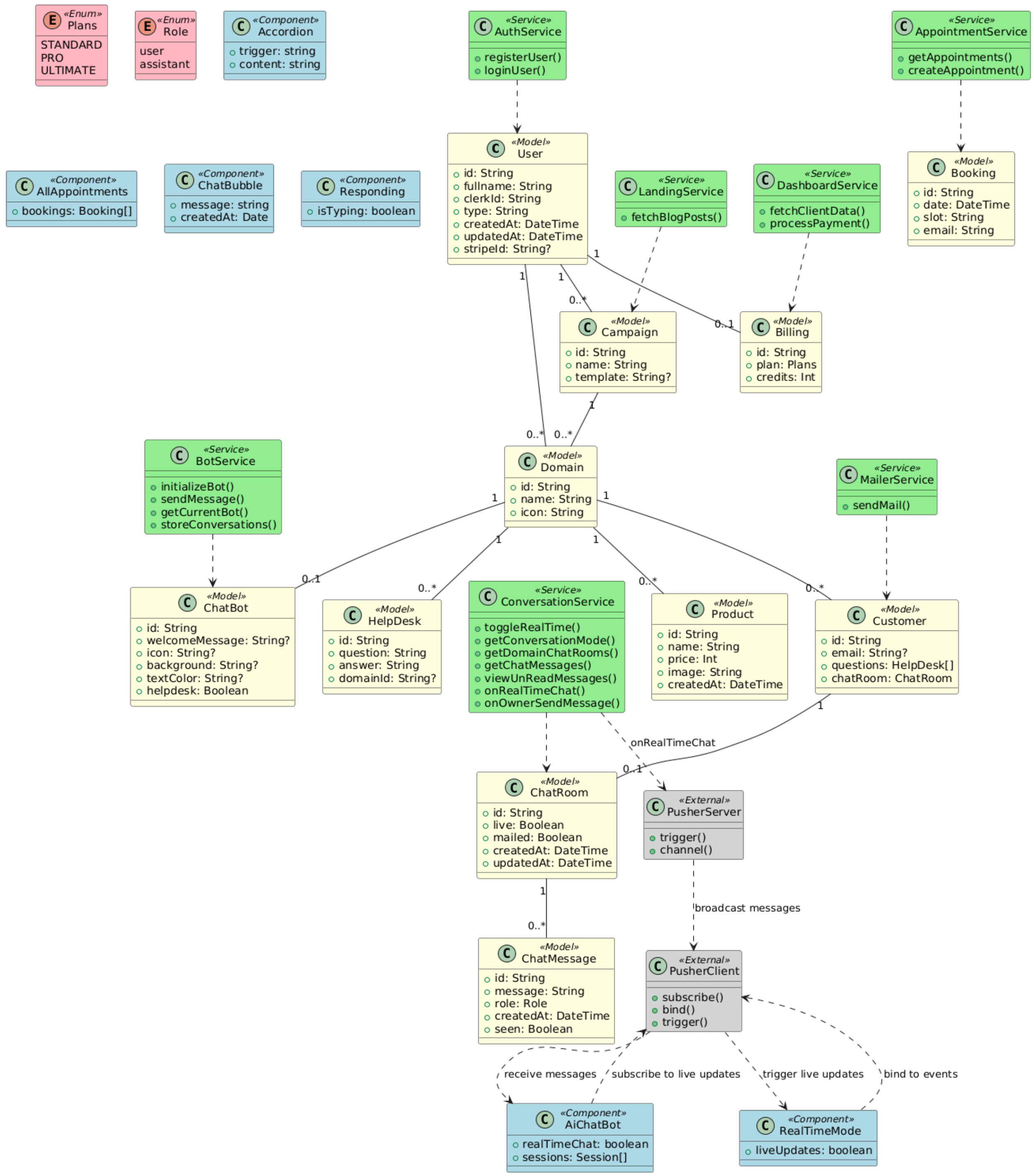

Figure 4.15 Class Diagram of Corinna Ai

Chapter 5

Implementation

5. Implementation

This chapter details the specific algorithms and implementation strategies used to build the core functionalities of Corinna AI, including the AI response logic, real-time handoff mechanism, and the user interface construction.

5.1. Algorithm

5.1.1. Prompt Engineering

The assistant is primed with a detailed role prompt, instructing it to behave like a professional sales rep for a specific brand aka domain.

CODE SNIPPET

```
messages: [
  { role: 'assistant',
    content: `
      You are a highly knowledgeable and experienced sales representative for a ${chatBotDomain.name} that offers a valuable product or service. Your goal is to have a natural, human-like conversation with the customer in order to understand their needs, provide relevant information, and ultimately guide them towards making a purchase or redirect them to a link if they haven't provided all relevant information.

        Right now you are talking to a customer for the first time. Start by giving them a warm welcome on behalf of ${chatBotDomain.name} and make them feel welcomed.

        Your next task is lead the conversation naturally to get the customers email address. Be respectful and never break character. `, }

 ...chat, { role: 'user', content: message,},
```

5.1.2. **Context-Based Switching**

The chatbot will be able to switches from AI to human/live agent based on customer behavior or message content by performing sentimental analysis.

- How to use the filter question array.
- Detects keyword (realtime) in AI response.
- If detected → toggles chatRoom.live = true to enable human operator.

CODE SNIPPET

```
messages: [
  { role: 'assistant',
    content: ` if the customer says something out of context or inappropriate. Simply say this is beyond you and you will get a real user to continue the conversation. And add a keyword (realtime) at the end. `,
  } ...chat, { role: 'user', content: message,},
```

5.1.3. **Dynamic Question Flow Control**

The purpose of this implementation is to automate follow-up questions using an array of filterQuestions, to track which ones are answered.

- Prompts GPT to ask from a provided list.
- After user responds, include the keyword `complete`at the end of the question.
- If so → mark the unanswered customerResponse as answered.

PSEUDOCODE

```
IF the last question is the chat contains the keyword "(complete)" THEN

  FIND the first unanswered question from the database

  IF an unanswered question is found THEN

    UPDATE the current message with the 'unanswered question'

  ENDIF

ENDIF
```

5.1.4. **Smart Onboarding & Email Extraction**

Detects customer email from the first message and auto-creates a Customer + ChatRoom entry if new.

- Uses extractEmailsFromString() to extract email from free-form text.
- Checks if email exists → if not, creates new customer entry and populates questions.

PSEUDOCODE

```
extractedEmail = extractEmailsFromString(message)

checkCustomer = find Customers WHERE email IN extractedEmail

IF checkCustomer is empty:

  UPDATE Domain SET: CREATE new Customer WITH:

        email = extractedEmail,

        questions = create from chatBotDomain.filterQuestions,

        chatRoom = create empty chatRoom
```

5.1.5. **Conversation Persistence**

The system saves every message exchanged to chatRoom.message with role and timestamp.

- Each time a message is sent or received, it's logged via onStoreConversations()

PSEUDOCODE

```
FUNCTION storeConversation(chatRoomId, messageText, senderRole):
  UPDATE ChatRoom WHERE id == chatRoomId SET:
    CREATE new Message WITH:
      message = messageText
      role = senderRole
```

5.2. **External APIs**

The Corinna AI platform integrates several specialized third-party APIs to handle complex functionalities such as artificial intelligence, real-time communication, and secure payments. This integration allows the system to remain lightweight and modular while leveraging enterprise-grade services.

Table 5.1 Details of APIs used in the project

Name of API	**Description of API**	**Purpose of usage**	**List down the function/class**
OpenAI API [24]	A cloud-based interface for accessing Large Language Models (LLMs) like GPT-3.5 and GPT-4.	Powering the core intelligence of the chatbot to generate natural, context-aware responses and perform role-based selling tasks.	**Backend:** lib/openai.ts, POST /api/conversation **Frontend:** useChatBot hook (triggering the AI response)

Stripe [25]	A comprehensive financial infrastructure platform for the internet.	Managing secure credit card transactions, processing monthly SaaS subscriptions, and handling billing webhooks.	**Backend:** api/stripe/webhook **Frontend:** onSubscribe function (Billing Dashboard)
Clerk	A complete user management and authentication suite built for the modern web.	Handling secure user sign-up, login, session management, and enforcing Role-Based Access Control (RBAC).	**Middleware:** middleware.ts **Frontend:** <ClerkProvider>, useAuth() hook
Pusher	A hosted real-time messaging service acting as a simplified WebSocket wrapper.	Facilitating the bi-directional, low-latency communication required for the **Real-Time Handoff** feature and instant chat updates.	**Frontend:** useRealTime hook (in real-time.tsx), PusherClient **Backend:** pusherServer.trigger()
Uploadcare	An end-to-end file handling platform and Content Delivery Network (CDN).	Storing and serving static assets such as chatbot icon images and PDF documents used for knowledge base training.	**Frontend:** <UploadButton/> component (in window.tsx) **Rendering:** bubble.tsx (displaying ucarecdn images)

5.3. User Interface

The user interface (UI) of Corinna AI is designed with a focus on usability, responsiveness, and aesthetic clarity. Built using **Next.js** and **React**, the interface leverages **Tailwind CSS** for consistent styling and utilizes component-based architecture to ensure modularity. The system also incorporates **Shadcn UI** components (such as Avatars, Cards, and Tabs) to maintain a polished, accessible, and professional design standard.

Landing Page

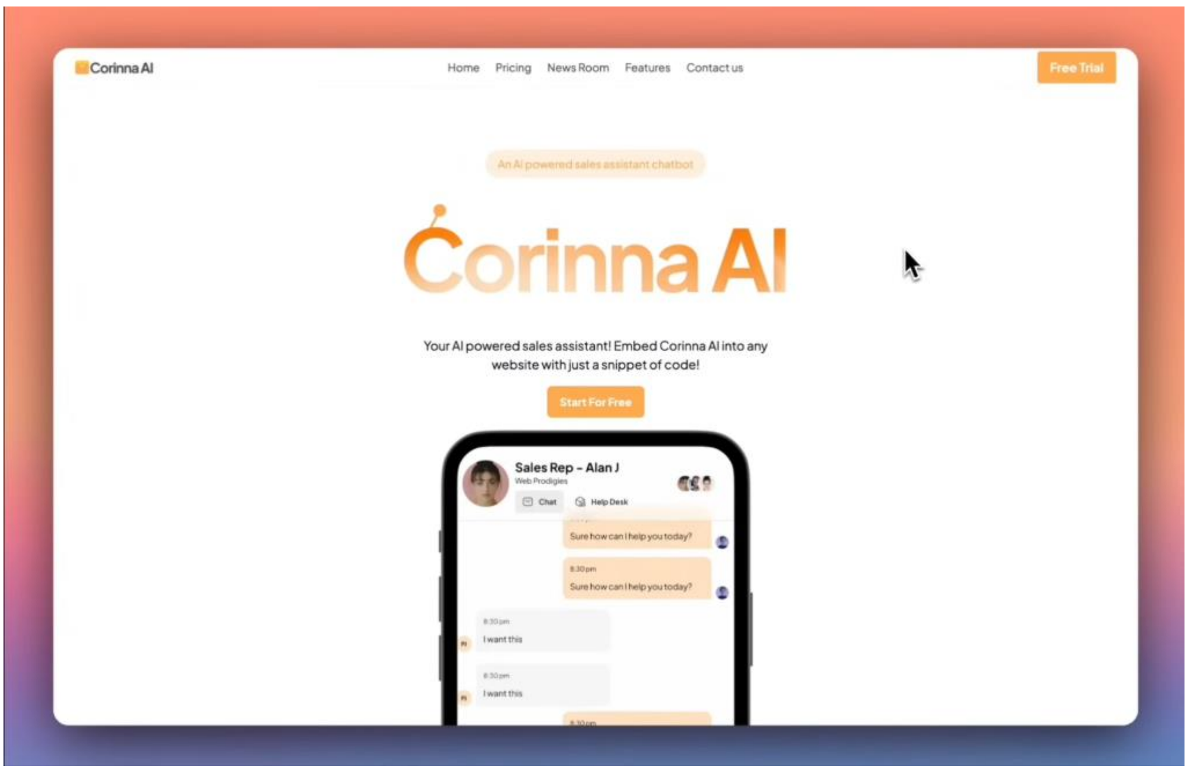

Figure 5.1 Landing Page

To drive user engagement and encourage immediate conversion, the landing page features two strategically placed Call-to-Action (CTA) buttons. A **"Free Trial"** button is positioned in the top-right corner of the header, serving as a persistent anchor that allows users to access the sign-up process from anywhere on the site. Additionally, a primary **"Start For Free"** button is centered prominently beneath the main value proposition; utilizing the brand's signature orange color, this button visually guides the user’s journey from the headline directly to the sign-up action, highlighting the platform's low barrier to entry.

Account Creation & Onboarding Interface

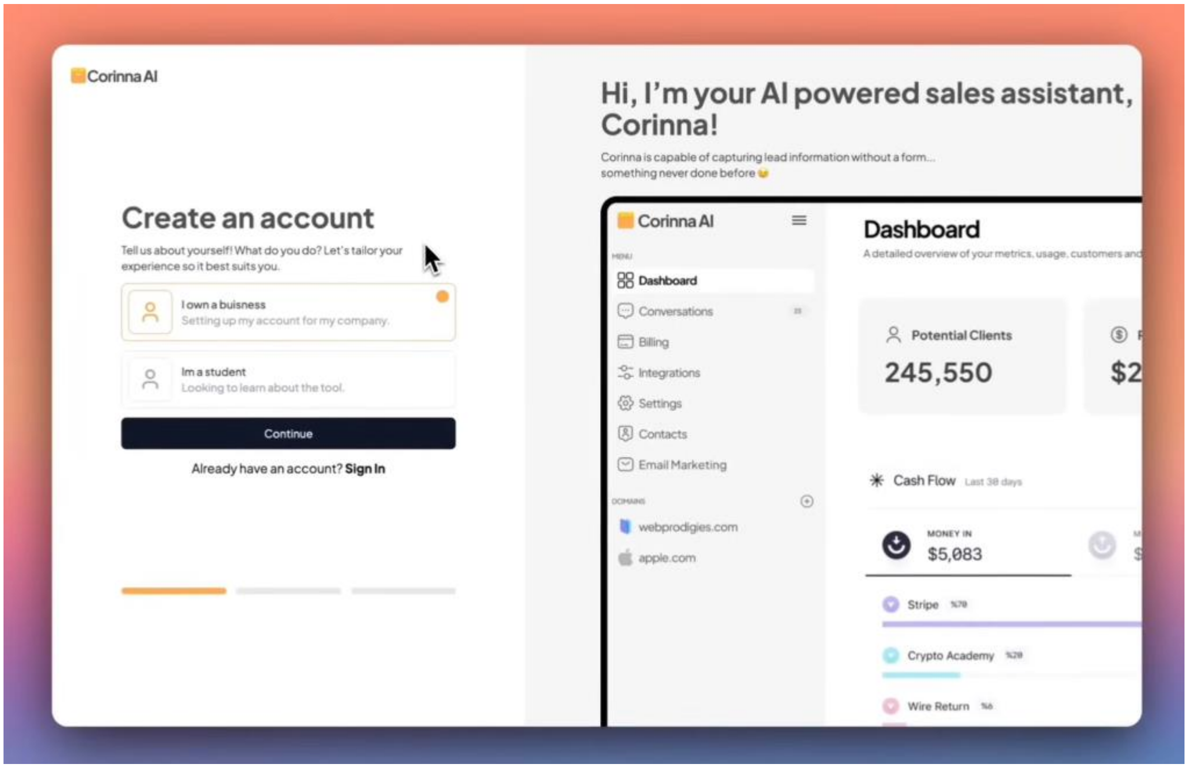

Figure 5.2 Onboarding Page

The account creation screen features a user-friendly split-layout design that guides users through a personalized onboarding process. The left side prompts users to identify their role—such as a business owner or student—using interactive selection cards to tailor the experience, while a bottom progress bar indicates the setup steps. Complementing this, the right panel displays a welcoming message from the AI assistant and a detailed preview of the application dashboard, offering an immediate visual of key features like client tracking and cash flow metrics to reinforce the product's value before the signup is complete.

User Dashboard & Analytics

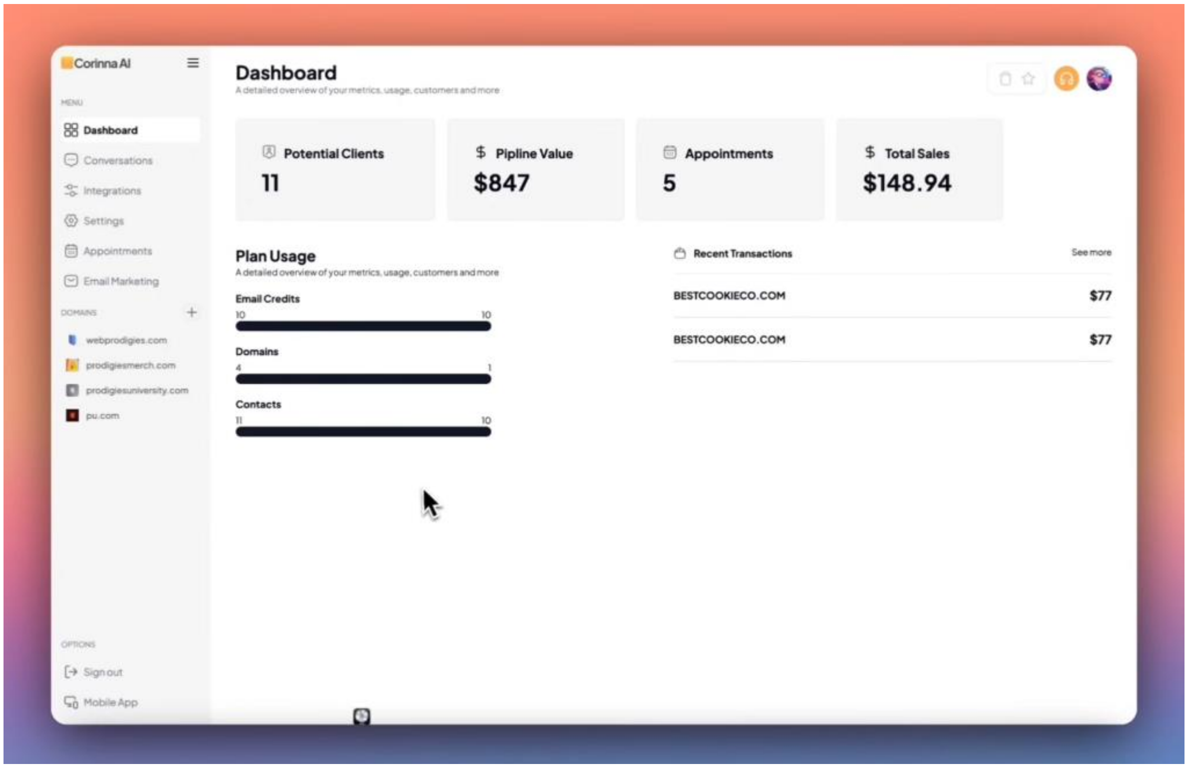

Figure 5.3 Dashboard Page

The Corinna AI Dashboard serves as the central command center for the user, featuring a clean sidebar navigation on the left for easy access to core modules such as Conversations, Integrations, and Domain settings. The main workspace is anchored by four prominent key performance indicator (KPI) cards that display real-time data for Potential Clients, Pipeline Value, Appointments, and Total Sales. Below these top-level metrics, the interface provides a detailed "Plan Usage" section, which utilizes visual progress bars to track resource limits like email credits and contacts, alongside a "Recent Transactions" panel that offers a quick log of the latest financial activity.

Domain Settings & Integration

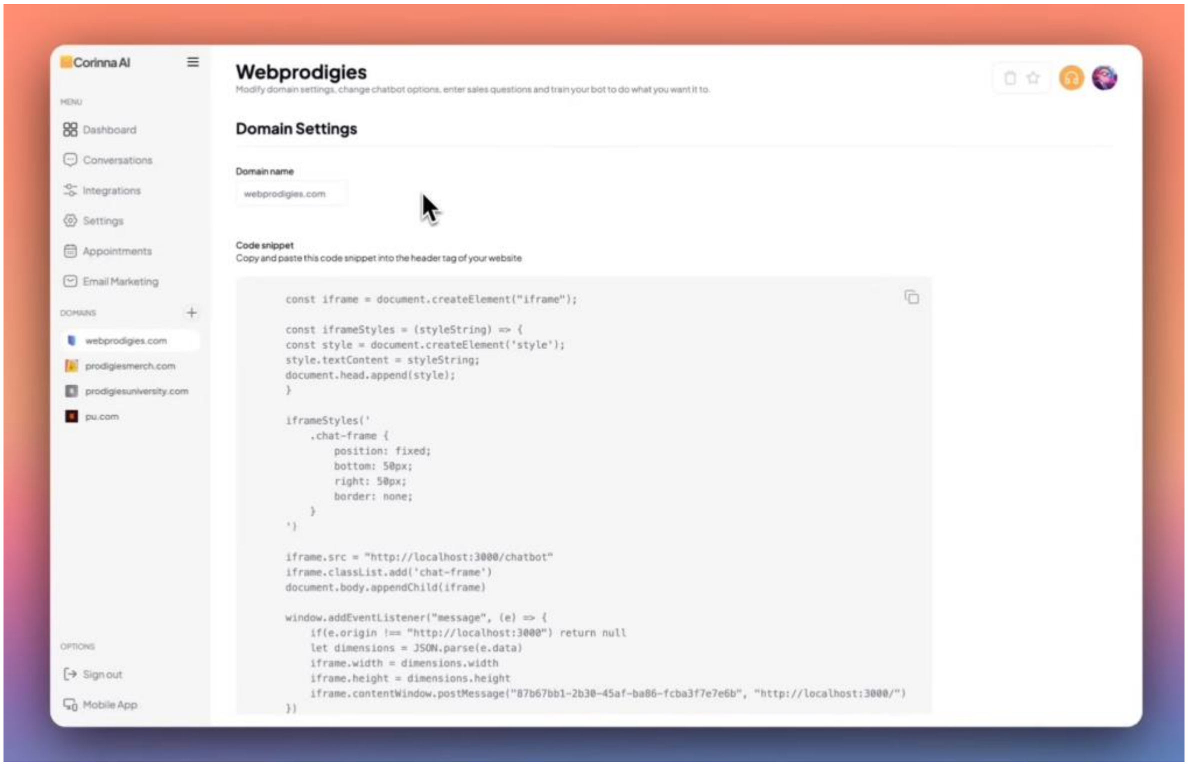

Figure 5.4 Domain Settings Page

The Domain Settings interface serves as the technical integration hub, allowing users to configure specific web properties and connect the chatbot to their live sites. The focal point of this screen is the "Code snippet" section, which provides a pre-generated block of JavaScript that users can simply copy and paste into their website's header tag to embed the chatbot instantly. The layout clearly displays the active domain (e.g., "Webprodigies") with editable fields and maintains context through the sidebar, ensuring users can easily switch between different projects while managing their integration scripts.

Bot Training & Help Desk Configuration

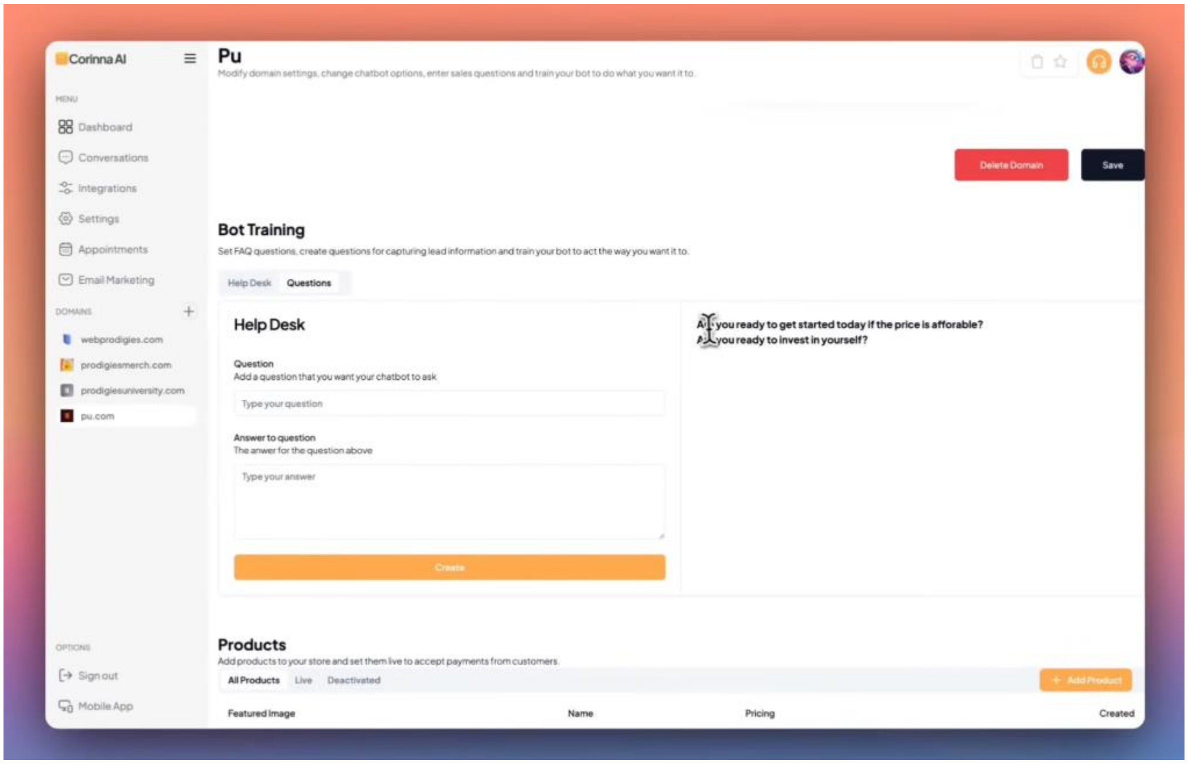

Figure 5.5 Bot Training & Help Desk Configuration

The Bot Training interface empowers users to customize their AI’s conversational behavior through a dedicated "Help Desk" configuration tool. The screen features a clear input form where administrators can define specific questions and answers, allowing them to script the chatbot for FAQs or lead qualification. Once added via the "Create" button, these interactions appear in a list on the right for easy management. Furthermore, a "Products" section at the bottom allows for the integration of inventory items, ensuring the bot is trained to discuss and sell specific offerings directly within the chat.

Third-Party Integrations

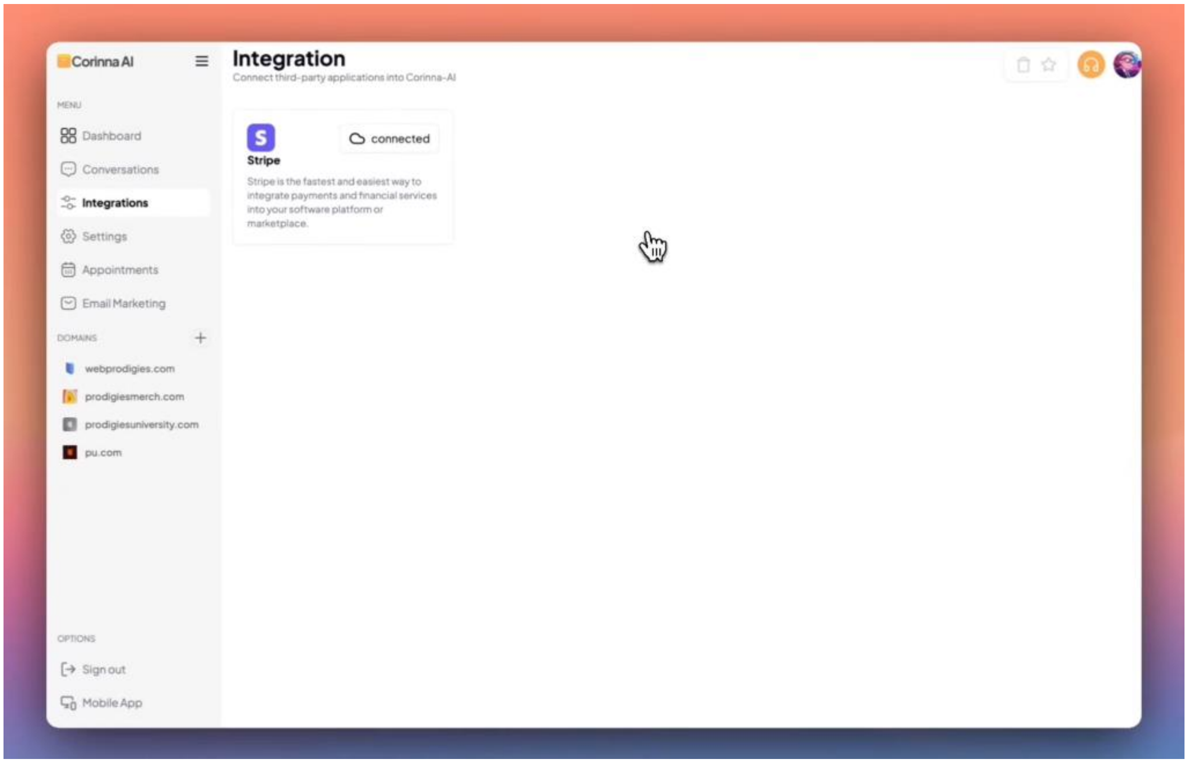

Figure 5.6 Third-Party Integrations

The Integration management screen provides a streamlined interface for connecting third-party applications to the Corinna AI ecosystem. The layout utilizes a modular card-based system to display available services, as demonstrated by the active **Stripe** integration card shown in the view. Each integration card offers a brief description of the service's functionality and features a clear status indicator (such as the "connected" badge), allowing users to easily monitor the state of their external connections for payments and financial services at a glance.

Account Settings & Personalization

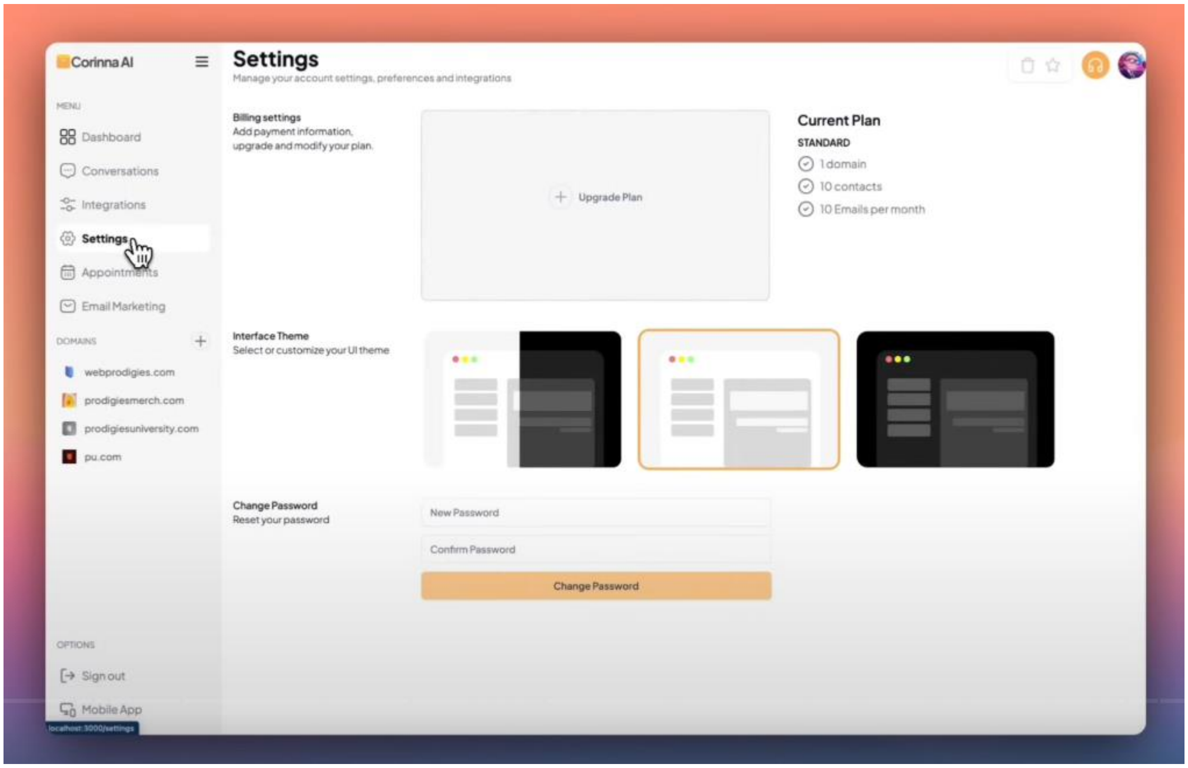

Figure 5.7 Account Settings & Personalization

The Settings interface functions as a comprehensive control panel for account management and personalization. The layout is divided into clear sections: "Billing settings" provides options for plan upgrades, while a "Current Plan" summary on the right offers a transparent view of active resource quotas, such as domain and contact limits. Users can also customize their visual experience through the "Interface Theme" selector—which displays interactive previews of light, dark, and split-mode options—while a dedicated security section at the bottom allows for immediate password updates.

Live Chatbot Widget

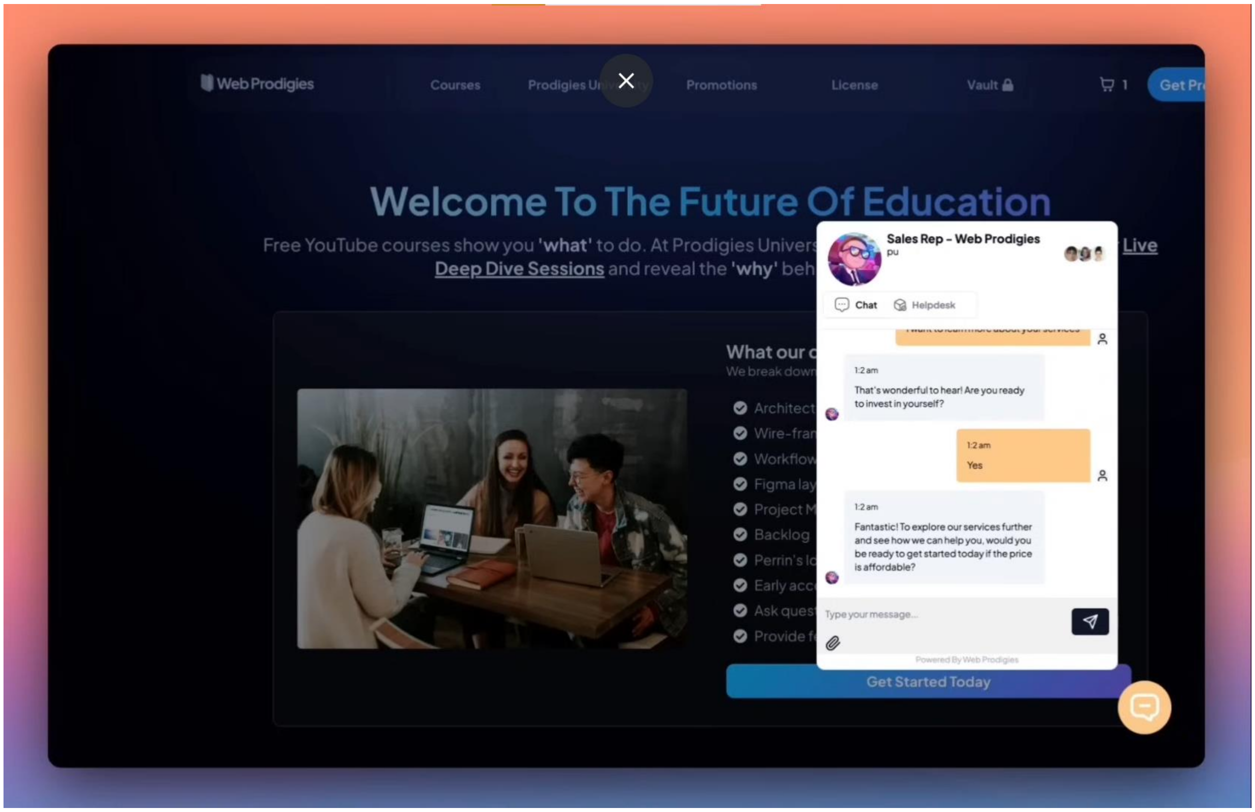

Figure 5.8 Live Chatbot Widget

The live Chatbot interface is designed as a sleek, non-intrusive floating widget that overlays the host website to provide immediate support. Visually, it features a personalized header identifying the "Sales Rep" to humanize the interaction, along with a dual-tab system that allows users to switch between "Chat" for real-time conversation and "Helpdesk" for self-service resources. The conversation window displays a clear, thread-based message history with distinct bubbles for the user and the AI, while the footer provides a streamlined input area equipped with file attachment capabilities and a send button, ensuring a seamless and modern communication experience for site visitors.

Lead Qualification Form

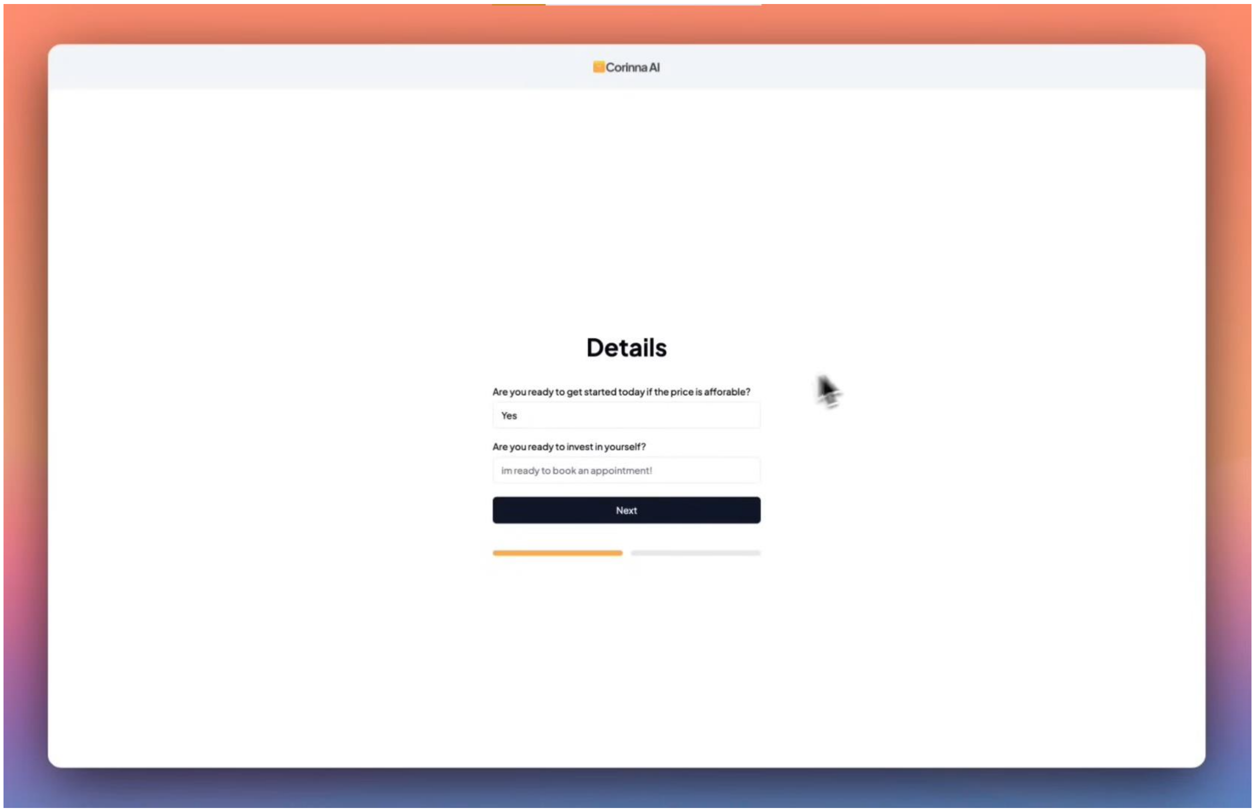

Figure 5.9 Lead Qualification Form

The "Details" interface functions as a streamlined data collection step designed to qualify user interest through specific, targeted questions. The minimalist layout centers on key prompts—asking about pricing sensitivity and investment readiness—followed by text input fields for custom responses. To ensure a smooth user experience, the screen utilizes a multi-step progress bar at the bottom and a prominent "Next" button, guiding the user clearly to the subsequent stage of the process.

Appointment Booking Interface

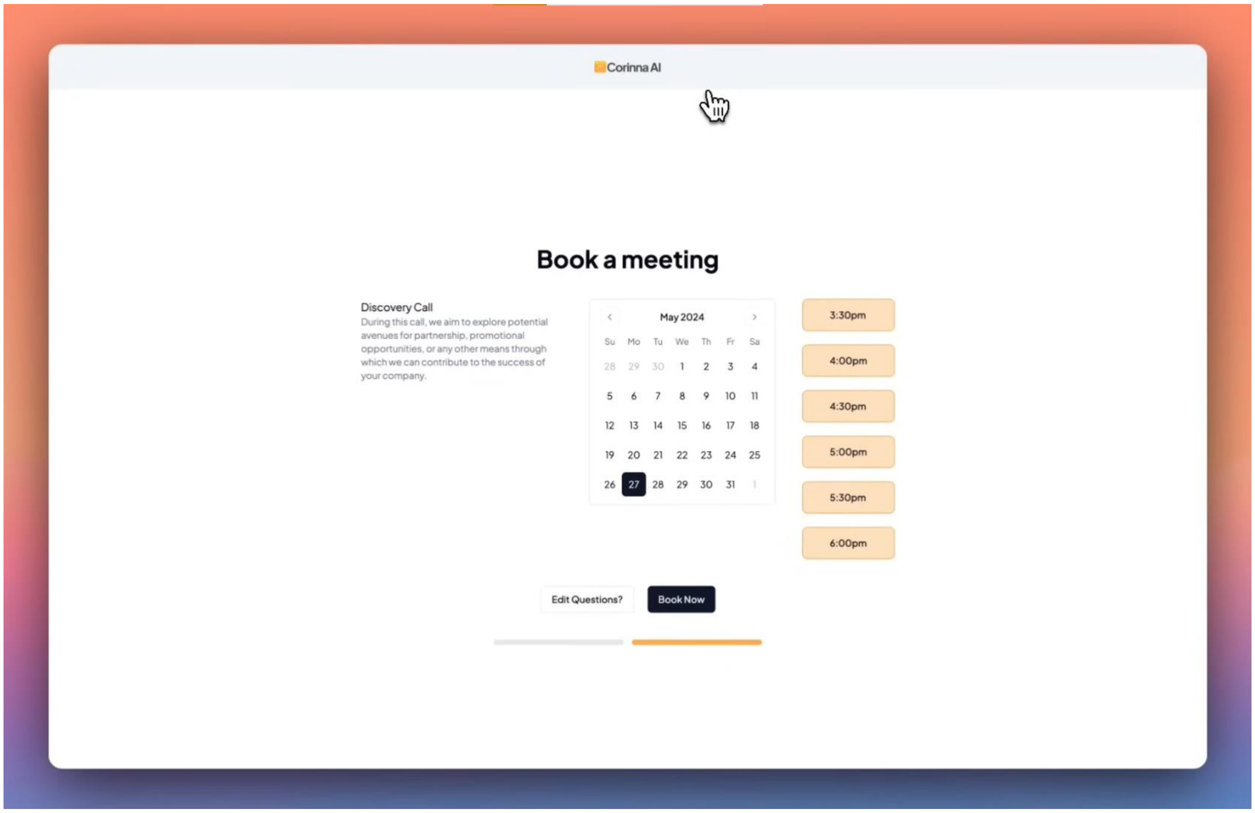

Figure 5.10 Appointment Booking Interface

The meeting booking interface streamlines the scheduling process by presenting a clean, user-friendly "Discovery Call" selector. Central to the layout is an interactive calendar widget allowing specific date selection, paired with a vertical list of available time slots on the right for immediate convenience. The screen is grounded by a clear primary action button labeled "Book Now" and a secondary option to "Edit Questions?", all sitting above a progress indicator that visually tracks the user's advancement through the multi-step scheduling workflow.

Secure Payment Gateway

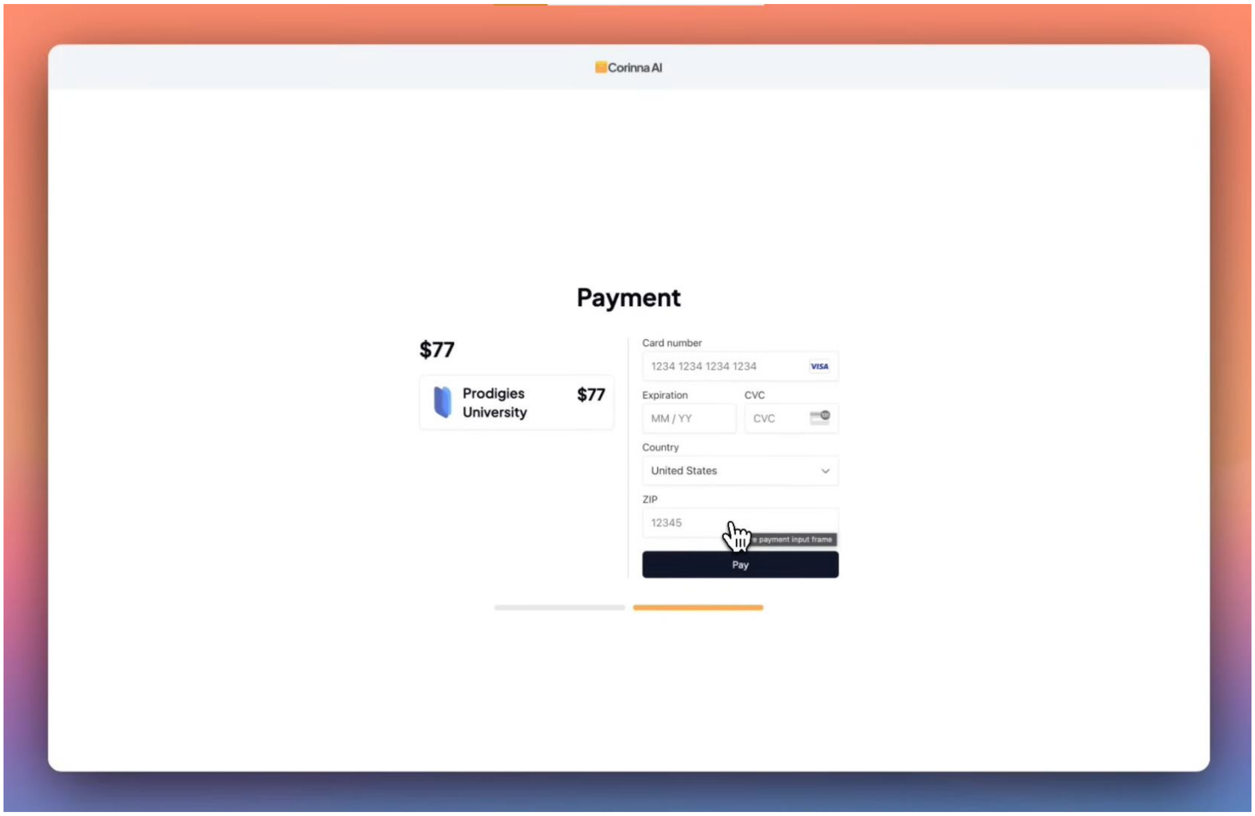

Figure 5.11 Secure Payment Gateway

The Payment interface is designed to provide a seamless and secure checkout experience, minimizing friction at the final conversion point. The minimalist layout clearly separates the order summary displaying the specific product ("Prodigies University") and price—on the left, while the right side presents a clean, standard data entry form for credit card details (Number, Expiration, CVC, and ZIP). The screen is grounded by a high-contrast, dark "Pay" button to drive the action, and a bottom progress bar visually confirms this as the final step in the transaction workflow, ensuring users feel oriented and confident in their purchase.

Real-Time Conversation Inbox

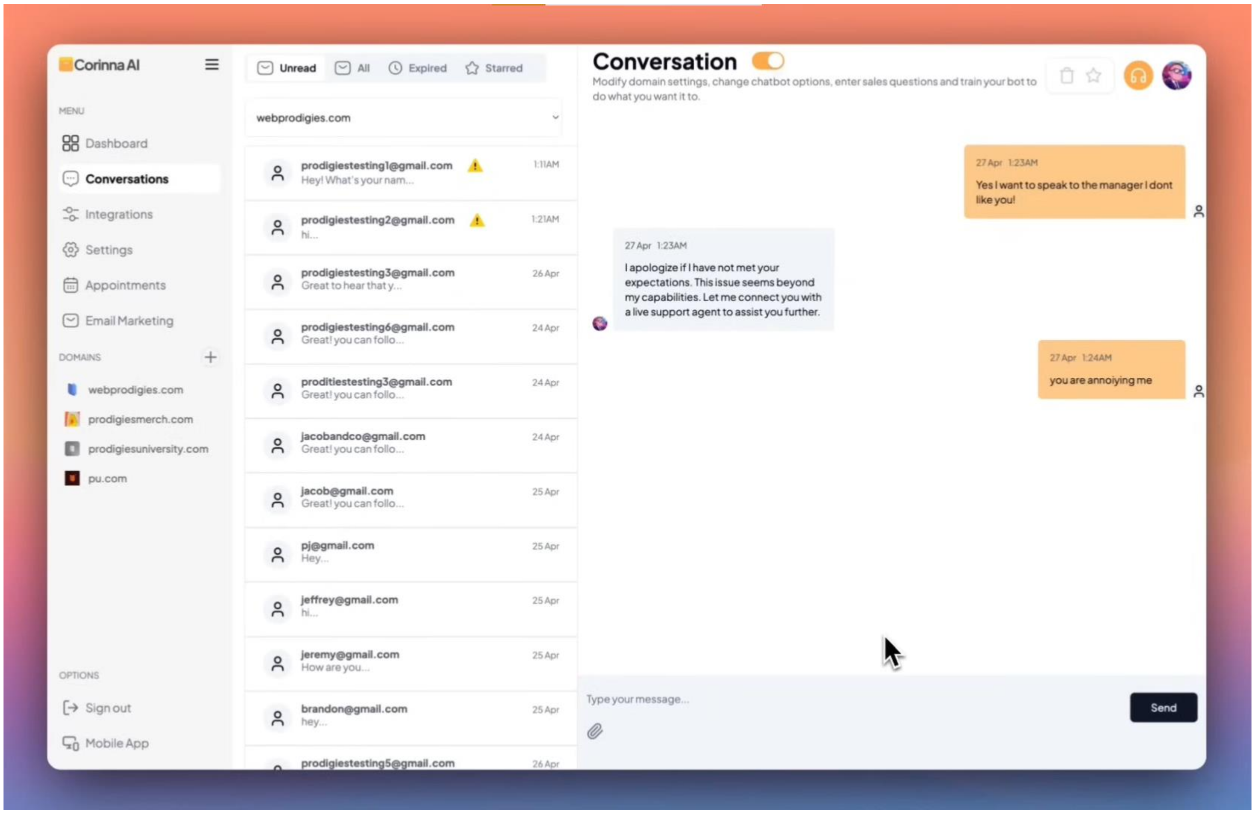

Figure 5.12 Real-Time Conversation Inbox

The Conversation management interface functions as a centralized hub for monitoring and intervening in user interactions. The layout features a productive three-panel design: the global navigation on the left, a detailed inbox list in the center—equipped with filters for "Unread," "All," and "Starred" messages to prioritize workflow—and the active chat window on the right. This main workspace displays a chronological message history with clear visual distinctions between user inquiries and bot responses, while the manual input area allows human agents to seamlessly take over the conversation when complex support is required.

Appointment Management System

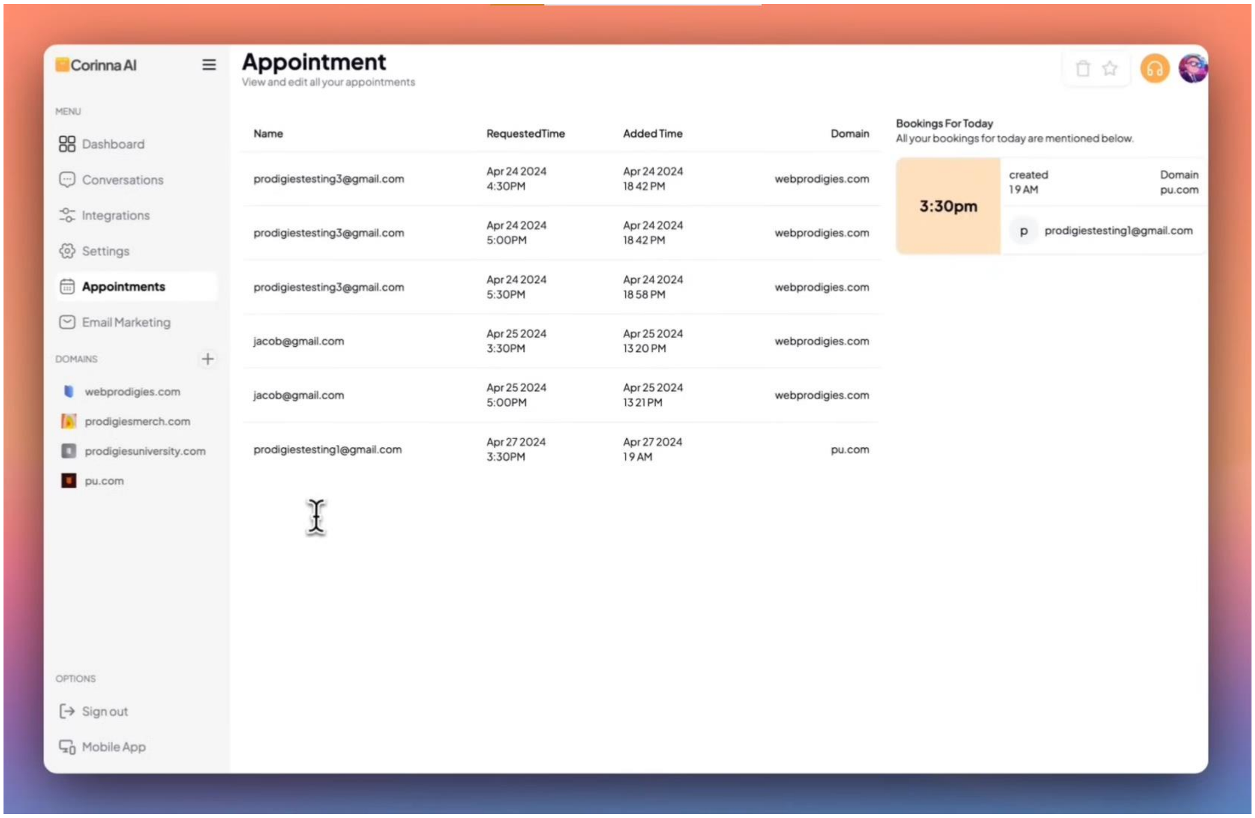

Figure 5.13 Appointments

The Appointment management screen offers a comprehensive, centralized view of all scheduled client interactions. The main workspace features a detailed data table that organizes appointments chronologically, displaying critical information such as client names, requested times, and the associated domain for each meeting. To assist with daily prioritization, a dedicated "Bookings For Today" panel on the right highlights immediate upcoming schedules in a high-visibility card, ensuring users never miss a critical sales call or lead engagement.

Email Marketing Campaign Manager

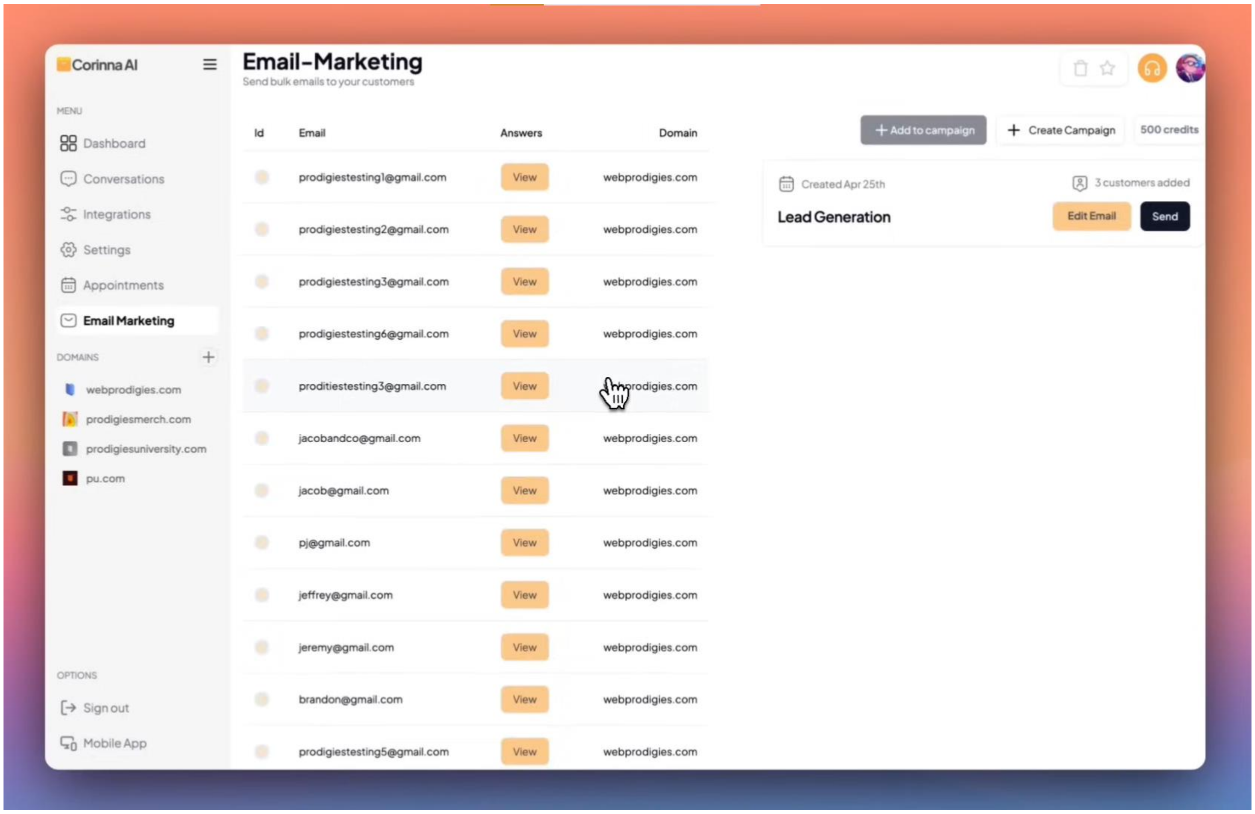

Figure 5.14 Email Marketing Campaign Manager

The Email Marketing interface functions as a targeted outreach hub, enabling users to manage and execute bulk communications directly with captured leads. The layout features a functional split design: the left side presents a detailed list of potential customers, displaying their email addresses and associated domains alongside a "View" button to access their specific chatbot responses. The right side focuses on active campaign management, allowing users to create new initiatives, monitor available sending credits (e.g., "500 credits"), and execute blasts—such as the "Lead Generation" campaign shown—via a high-contrast "Send" button.

Chapter 6

Testing and Evaluation

6. Testing and Evaluation

This chapter outlines the testing methodologies applied to the **Corinna AI** platform to ensure its correctness, stability, and performance under real-world SaaS conditions. A blend of manual and automated testing strategies was used to verify the integrity of key modules such as user authentication, real-time chat synchronization (WebSockets), AI response generation, and payment processing. Testing was conducted in both development and production-like environments to ensure Corinna AI handles edge cases, API failures, and latency-sensitive operations gracefully.

Special attention was given to the correctness of the **AI-to-Human Handoff**, handling of asynchronous events via Pusher, resilience of the OpenAI API integration, and accuracy of lead capture calculations. Real-time scenarios, such as network disconnections, invalid user inputs, and concurrent chat sessions, were simulated to assess system fault tolerance. The platform's logging and alerting mechanisms were tested to confirm that anomalies are properly recorded and actionable notifications are triggered on the Agent Dashboard.

6.1. Manual Testing

Manual testing was employed to validate user workflows, UI behavior, and backend operations such as account registration, chatbot widget embedding, live chat toggling, and appointment booking. This allowed testers to simulate real business owner and website visitor interactions and assess whether the system's behavior aligned with business logic and expected outcomes. Critical functionalities such as failed payment processing, invalid prompt injections, and status monitoring were manually verified under different scenarios [26].

Additionally, the manual testing process included testing various user permission levels (e.g., Trial User vs. Pro User) and verifying access control restrictions across modules. Testers ensured that error boundaries were gracefully handled—for example, when the OpenAI API times out or when a user attempts to access a protected route without a valid session.

6.1.1. **System Testing**

In system testing, the entire Corinna AI application—including the Next.js frontend, API backend, Neon database, and external integrations (Stripe, OpenAI, Pusher)—was evaluated as a complete, integrated system. The goal was to confirm that the full-stack application fulfilled all functional requirements. Testers executed end-to-end flows such as:

- **Visitor Journey:** Landing on a client site -> Opening widget -> Chatting with AI -> Switching to Human -> Booking a Meeting.
- **Agent Journey:** Logging in -> Configuring Chatbot -> Receiving Real-time Alert -> Taking over chat -> Managing Lead.

Stress tests were simulated by triggering multiple concurrent chat sessions to observe how the serverless functions and database connections handled increased load. Integration points between modules were closely monitored to detect latency bottlenecks (e.g., slow AI responses) or missed synchronization events (e.g., dashboard not updating).

6.1.2. **Unit Testing**

Unit testing focused on verifying individual logic blocks and utility functions involved in tasks such as email validation, password strength checks, prompt construction, and date formatting. These tests ensured that critical algorithms behaved correctly in isolation.

Unit Testing 1: Signup Testing Objective: Verify successful account creation via Clerk and ensure all validation rules are correctly applied during the signup process.

Table 6.1 Unit Testing – Signup Scenarios

No.	Test Case / Script	Attribute and Value	Expected Result	Result
1	Valid signup credentials	**Email:** newuser@corinna.ai **Password:** Test@123	Account created in Clerk and Database, user	**Pass**

			redirected to onboarding.	
2	Invalid email format	**Email:** userexample.com **Password:** Test@123	Error message: “Invalid Email Format” displayed on form.	**Pass**
3	Empty email field	**Email:** (empty) **Password:** Test@123	Error message: “Email field is required”	**Pass**
4	Weak password	**Email:** user@corinna.ai **Password:** 123	Error message: “Password must be at least 8 characters”	**Pass**
5	Passwords do not match	**Password:** Test@123 **Confirm:** Test@124	Error message: “Passwords do not match”	**Pass**
6	Email already registered	**Email:** existing@corinna.ai **Password:** Test@123	Error message: “Email already in use”	**Pass**

Unit Testing 2: Login Testing Objective: Test the authentication process, verify correct error handling for invalid credentials, and ensure proper login functionality.

Table 6.2 Unit Testing – Login Scenarios

No.	Test Case / Script	Attribute and Value	Expected Result	Result
1	Valid login credentials	**Email:** owner@corinna.ai **Password:** Test@123	Successful login, redirected to /dashboard	**Pass**

2	Invalid email format	**Email:** user@ **Password:** Test@123	Error message: “Invalid Email Format”	**Pass**
3	Empty fields	**Email:** (empty)	Error message: “All fields are required”	**Pass**
4	User not found	**Email:** ghost@corinna.ai	Error message: “User not found” or “Invalid Credentials”	**Pass**
5	Incorrect password	**Email:** owner@corinna.ai **Password:** wrongpass	Error message: “Invalid password”	**Pass**

Unit Testing 3: Chatbot Configuration Testing Objective: Validate the process of updating chatbot settings (Name, Welcome Message), ensuring proper persistence in the Neon database [27].

Table 6.3 Unit Testing – Chatbot Config Scenarios

No.	**Test Case / Script**	**Attribute and Value**	**Expected Result**	**Result**
1	Valid configuration update	**Name:** "Sales Bot" **Message:** "Hi there!"	Settings updated successfully, success notification shown.	**Pass**
2	Missing required fields	**Name:** (empty)	Error message: “Chatbot Name is required.”	**Pass**

3	Invalid file upload	**File:** video.mp4 (instead of PDF/Txt)	Error message: “Invalid file type. Only PDF/TXT allowed.”	**Pass**
4	Oversized file upload	**File:** large_book.pdf (>10MB)	Error message: “File size exceeds limit.”	**Pass**

Unit Testing 4: AI Response Logic Testing Objective: Verify that the backend correctly constructs prompts and handles OpenAI responses.

Table 6.4 Unit Testing – AI Logic Scenarios

No.	Test Case / Script	Attribute and Value	Expected Result	Result
1	Valid User Query	**Input:** "What is the price?"	AI returns a relevant response based on context.	**Pass**
2	Empty User Query	**Input:** ""	Backend returns 400 Bad Request error.	**Pass**
3	Context Keyword Trigger	**Input:** "I want to speak to a human"	AI response includes (realtime) keyword.	**Pass**
4	API Failure Simulation	**OpenAI:** 500 Error	Backend catches error, returns "Service unavailable" message.	**Pass**

6.1.3. **Functional Testing**

Functional testing focuses on ensuring that the system performs its intended operations accurately and consistently, as outlined in the functional specifications [28].

Functional Testing 1: User Roles and Dashboard Access Objective: Ensure that different users see the correct data and navigation options.

Table 6.5 Functional Testing – Role Access

No.	Test Case	Precondition	Steps	Expected Result	Status
1	Business Owner Login	Valid Account	1. Navigate to Login 2. Enter Owner creds	Dashboard loads with "Settings", "Integrations", "Leads" tabs.	**Pass**
2	Website Visitor Access	Widget Embedded	1. Visit client site 2. Click Widget Icon	Chat window opens with "Chat" and "Help Desk" tabs visible.	**Pass**

Functional Testing 2: Real-Time Chat Handoff Objective: Verify the transition from AI mode to Live Agent mode.

Table 6.6 Functional Testing – Handoff Mechanics

No.	Test Case	Precondition	Steps	Expected Result	Status
1	Automatic AI Handoff	Chat Active	1. Visitor asks complex question	System toggles mode to live. Dashboard shows alert.	**Pass**

			2. AI detects urgency	Widget shows "Real Time" badge.	
2	Manual Toggle	Dashboard Open	1. Agent clicks "Real Time" toggle	Mode updates instantly on Visitor widget via Pusher.	**Pass**
3	Message Sync	Live Mode Active	1. Agent types "Hello" 2. Visitor types "Hi"	Both messages appear instantly on respective screens.	**Pass**

Functional Testing 3: Appointment Scheduling Objective: Verify the calendar booking flow.

Table 6.7 Functional Testing – Booking Flow

No.	**Test Case**	**Precondition**	**Steps**	**Expected Result**	**Status**
1	View Calendar	Chat Active	1. Visitor clicks "Book Meeting"	Calendar UI renders with available slots.	**Pass**
2	Confirm Slot	Slot Selected	1. Select Time 2. Confirm	Appointment saved to DB. Confirmation message displayed.	**Pass**

6.1.4. **Integration Testing**

Integration testing combines individual units to verify their interactions and integration points, particularly with external APIs.

Integration Testing 1: Payment and Subscription Objective: Ensure Stripe webhooks correctly update the application database.

Table 6.8 Integration Testing – Stripe & Database

No.	Test Case	Precondition	Steps	Expected Result	Status
1	Plan Upgrade	Free Plan User	1. User selects "Pro" plan 2. Completes payment on Stripe	Stripe sends webhook. Backend receives it and updates User.plan to PRO in Neon DB.	**Pass**
2	Payment Failure	Invalid Card	1. User inputs bad card number	Stripe rejects payment. UI shows error. Database remains on FREE plan.	**Pass**

Integration Testing 2: Email Marketing Trigger Objective: Ensure capturing a lead triggers the email engine.

Table 6.9 Integration Testing – Chat to Email

No.	Test Case	Precondition	Steps	Expected Result	Status
1	Capture Lead Email	Chat Active	1. Visitor provides email "test@mail.com"	Email saved to DB. Marketing Module triggers "Welcome	**Pass**

				Email" to "test@mail.com".	

6.1.5. **Automated Testing**

To ensure the application delivers an optimal user experience and adheres to web standards, automated auditing was performed using Google Lighthouse. This strategy focused on validating frontend performance metrics, accessibility standards, and SEO best practices to detect bottlenecks (e.g., render-blocking resources, unoptimized images) and ensure rapid page load times.

Table 6.10 Tools used-Automated Testing

Tool Name	**Tool Description**	**Applied on [list of related test cases / FR / NFR]**	**Results**
Google Lighthouse	Open-Source Automated Auditing Tool	**NFR-1 (Performance):** Used to audit page load metrics (FCP, LCP) and responsiveness.	**Performance Score: 99/100**

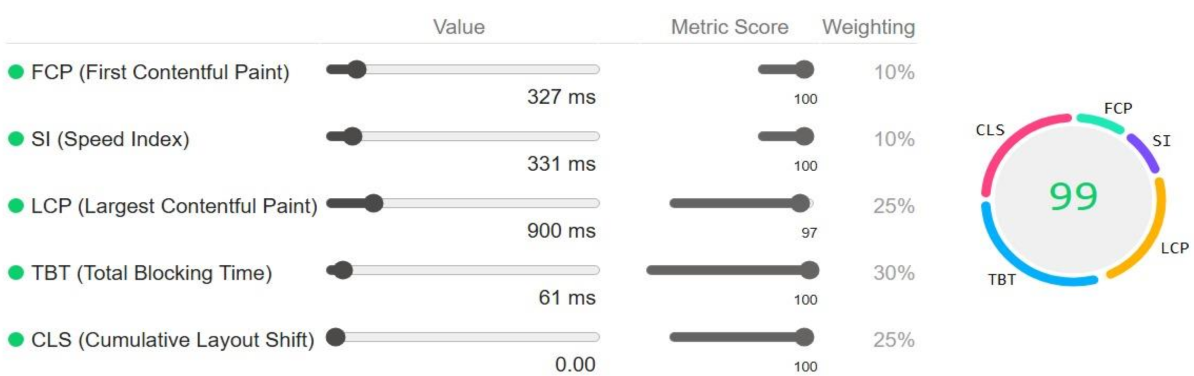

Figure 6.1 Lighthouse Testing Report

Chapter 7

Conclusion and Future Work

7. Conclusion and Future Work

This chapter concludes the project report by summarizing the achievements of Corinna AI against its initial objectives and outlining potential avenues for future development.

7.1. Conclusion

In conclusion, **Corinna AI** has successfully delivered a powerful, AI-driven Software as a Service (SaaS) platform designed to democratize sales automation for Small to Medium-sized Businesses (SMBs). By integrating cutting-edge technologies—specifically **Next.js** for full-stack architecture, **OpenAI** for generative intelligence, **Pusher** for real-time communication, and **Stripe** for financial infrastructure—the project offers a seamless, unified experience that replaces fragmented legacy tools.

The project achieved its primary objectives:

1. **Unified Workflow:** Successfully consolidated AI chat, live agent intervention, appointment scheduling, and email marketing into a single Agent Dashboard [29].
2. **Intelligent Handoff:** Implemented a novel **Context-Based Switching** mechanism that allows the AI to detect complex queries and trigger an instantaneous human takeover, significantly reducing lead leakage compared to standard chatbots [30].
3. **Scalable Architecture:** The adoption of a **Modular Monolith** design on a serverless stack (Vercel/Neon) proved effective, balancing rapid development velocity with the ability to scale resources dynamically.

Ultimately, Corinna AI empowers business owners to automate the initial, repetitive stages of the sales funnel while ensuring that high-value interactions are handled personally and instantly. The platform stands as a robust, cost-effective solution that bridges the gap between expensive enterprise CRM suites and basic, disconnected tools.

7.2. **Future Work**

While the current release of Corinna AI is a fully functional product, there are several strategic opportunities for future enhancement:

1. **Mobile Application for Agents:** Developing a native mobile app (iOS/Android) to allow business owners to receive push notifications and handle "Real-Time" chats on the go, rather than relying solely on the desktop dashboard.

2. **Omnichannel Support:** Expanding the chatbot's reach beyond the website widget to integrate with popular messaging platforms like **WhatsApp Business**, **Facebook Messenger**, and **Telegram**, allowing businesses to manage all communication channels from one inbox [31].

3. **Voice-Enabled Interaction:** Integrating voice-to-text and text-to-speech capabilities (e.g., OpenAI Whisper) to allow visitors to interact with the sales bot verbally, enhancing accessibility.

4. **Advanced Sentiment Analytics:** Implementing a dedicated analytics module that visualizes customer sentiment trends over time, helping businesses identify common pain points in their products or services.

5. **Automated Knowledge Base Scraping:** Enhancing the training module to automatically crawl and scrape a client’s website for product information, removing the need to manually upload PDF documents for training.

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